Fitting Species HT-DBH Curves Using FIA Data

A test with Douglas-fir in Oregon (Draft 1)

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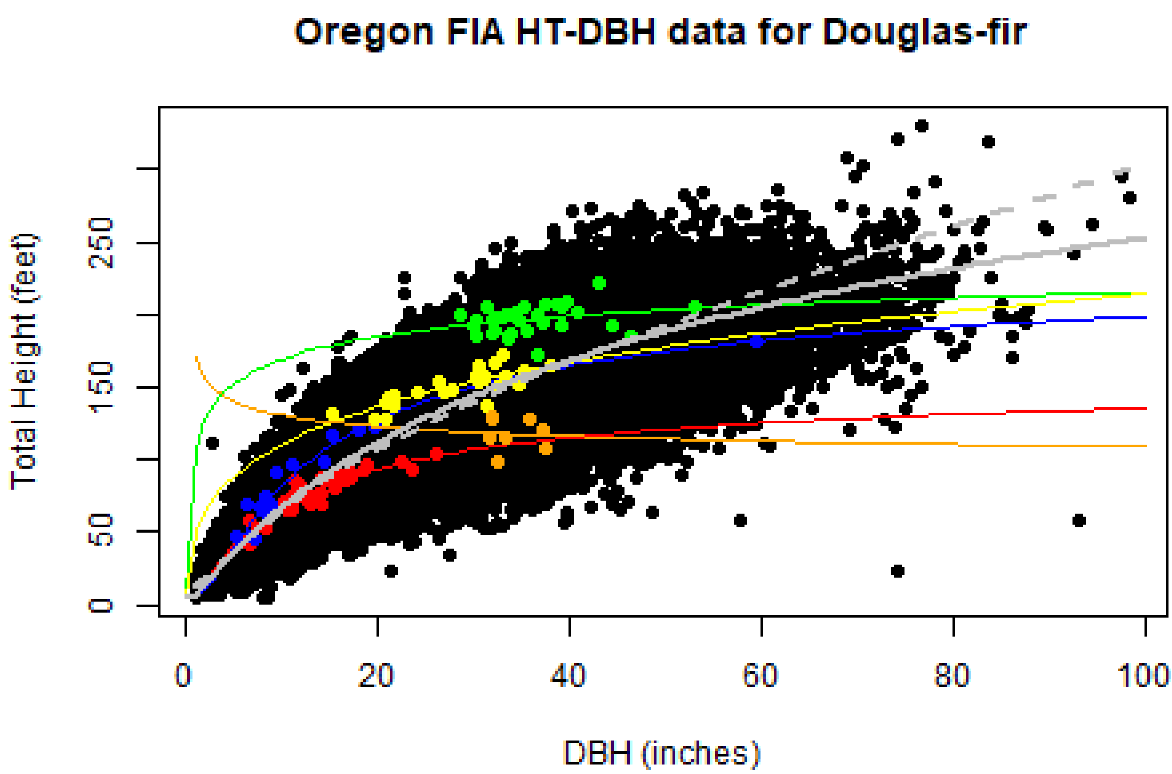
1.0 Introduction

Equations to predict tree total height (HT) from diameter at breast height (DBH) are a component of most tree-list type growth models (e.g., Wykoff and others 1982). These equations are used to impute missing heights and in some cases predict tree height growth from a tree's DBH at the start and end of the growth period. When applied in growth models, HT-DBH equations are typically fit to regional data for a species rather than individual stands. Unfortunately this causes some problematic behavior as demonstrated in the figure below which plots the measured heights for Douglas-fir from Forest Inventory and Analysis (FIA) plots across Oregon.



This graph also highlights the data from several individual FIA plots that represent different stages of stand development. Curves that were fit to these individual plots and the regional data are shown.

The graph shows some of the challenges with fitting and using regional HT-DBH curves:

1. The gray lines represent a HT-DBH curve fit to the regional data (solid line) and the lowess line (dashed). We might ask how the equation used to represent the relationship of HT to DBH characterizes the data for small and large DBHs and what is the appropriate asymptote?
2. The colored dots and associated lines for the measured Douglas-fir on individual plots across a range of stages of stand development as represented by the height of the 40-largest DBH trees per acre (HT40). These tend to be much flatter (different asymtotes) than the regional curve. Using a regional HT-DBH curve can give poor estimates for predicted heights for individual stands.
3. The plot represented by the orange color shows that fitting curves to individual stands can be challenging with small numbers of trees that represent a small range in tree sizes.

The work reported here is an exploratory analysis of fitting species specific HT-DBH curves across the species range using (FIA) data. It is part of a larger project that is working on developing a nation wide version of FVS that does not rely on regional variants. The purpose here is not to consider which is the best HT-DBH model but to look the potential impact of using regional models (ranther than stand-level models) and compare methods to localize them.

2.0 Methods

This analysis uses FAI Douglas-fir data for the state of Oregon which has 18818 plots, 525155 total trees records (trees and measurements) and 165469Douglas-fir trees measurements (some plots have remeasurements).

The full data set was filtered for just Douglas-fir (SPCD = 2020) with measured DBH (DIA, inches) and measured total HT (feet). Trees with a measured height had values of HT and an HTCD equal to 1 (field measured). Finally the data was filtered for plots with at least six measured Douglas-fir. This is not necessary for fitting regional HT-DBH equations, but was done here to be able to compare fits of individual FIA plots. The choice of six trees is an arbitrary minimum. This provided 6745 plots and 121335 Douglas-fir trees and measurements (some plots have remeasurements) which are shown in the figure above and summarized below.

Table 1: Filtered modeling data summary.

DIA	HT
Min. : 1.00	Min. : 5.00
1st Qu.: 8.10	1st Qu.: 54.00
Median :13.00	Median : 82.00
Mean :18.01	Mean : 94.09
3rd Qu.:26.20	3rd Qu.:129.00
Max. :94.40	Max. :331.00

There are many papers that have compared different HT-DBH equation for stands and regional models. The purpose of this work is not to evaluate model forms but to consider how best to use regional models to predict better for specific locations. Temesgen and others (2007) evaluated four widely used nonlinear HT-DBH models for 10 species using regional data from southwest Oregon (including Douglas-fir). They found the following exponential type curve to perform well across all the species and will be used for this analysis (for those interested, the four models were fit to the data set and are compared in the Appendix). Temesgen and van Gadow (2004) did a similar analysis of 8 species in British Columbia and also found this model to perform well (see tables 2 and 3).

$$HT = bh + exp(B_0 + B_1 * DBH^{B_2})$$

)

where bh is the breast height (4.5 feet) and the parameters are B0 > 0, B1 < 0 and B2 < 0. This will be refereed to as the Schumacher type model (Schumacher 1939).

There have been several regional HT-DBH models fit for Douglas-fir using different data sets using the above model form. The parameter estimates from three published examples of regional fits to Douglas-fir using this equation are given in the table below.

Table 2: Example fitted parameters for Douglas-fir in western Oregon and Washington.

Region	B0	B1	B2	Source
SE Oregon	7.128	-5.364	-0.2617	Larsen and Hann 1987
Willamette Valley	7.045	-5.168	-0.2539	Wang and Hann 1988
Western OR & WA	7.262	-5.9	-0.2872	Hanus and Others 1999

The above equation was fit to (1) all individual stands and measurements, (2) all data using nonlinear regression which was then calibrated to individual plots and (3) all data using a random effects model with the random effect on plot. Most plots have two measurements (usually about 10-years apart and a few plots have one or three measurements (plots with multiple measurements are considered "separate" plots for this analysis). For these fits no weights are applied (Martin and Flewelling 1998; Larsen and Hann 1987). Fitting was done using the R functions nls and nlsLM (minpack.lm package) for nonlinear least squares and nlme (nlme package). More specifics on fitting methods will be discussed in each section below.

Fits were compared with residuals plots, mean bias and root mean square error (RMSE).

$$BIAS = \sum(H_i - \hat{H}_i)^2$$

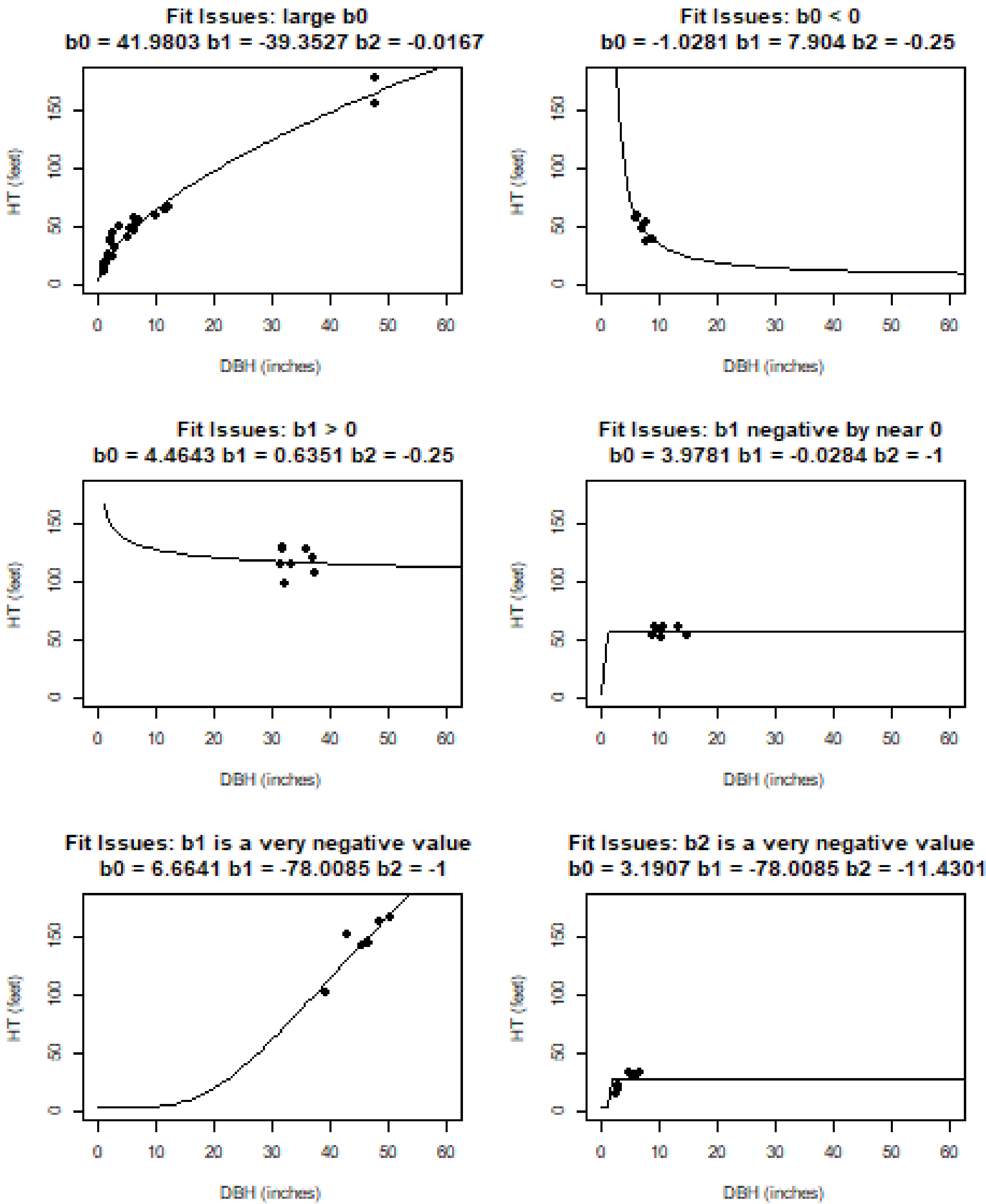
$$RMSE = \sqrt{\sum(H_i - \hat{H}_i)^2/N}$$

where Hi = measured tree heights and Hi hat is predicted tree heights. Note that these are summed across all trees on all plots/measurements which adds some complication since (a) not all plots have the same number of trees and (b) plots can have multiple measurements introducing correlations that are ignored. Fortunately, all comparisons use the same plots, measurements and trees.

3.0 Results

3.1 Fit All Plots Separately

As a "best case", tree heights were predicted by fitting individual HT-DBH curves for each FIA plot with at least six Douglas-fir with measured heights. The data has a wide range in conditions (numbers of trees and distribution of trees across the DBH range) that require a robust fitting procedure for reasonable plot level parameter estimates (see the discussion in Martin and Flewelling 1998). The following are some examples of initial fits that demonstrate some of the challenge for fitting individual plots and the issues to be addressed in fitting:



In order to fit curve that have reasonable behavior, the following steps were taken to fit HT-DBH curves to each individual plot:

step 1: To get a starting value for a nonlinear fit the equation was linearized by fixing the B2 parameter.

$$ln(HT - bh) = B_0 + B_1 * DBH^{-1.0}$$

Setting B2 = -1.0 seems to work well, but fitting a range of B2 values fixed to -0.25, -0.5, -0.75, -1.00, -1.25, -1.50, and -1.75 and pick the B2 with lowest RSE value is also an option. Below is an example for the range of B2 values fit to a sample plot. Lappi (1997) suggested that B2 could vary by age with values less than -1 for older stands and greater than -1 for younger stands. Martin and Flewelling (1998) came to a similar conclusion.

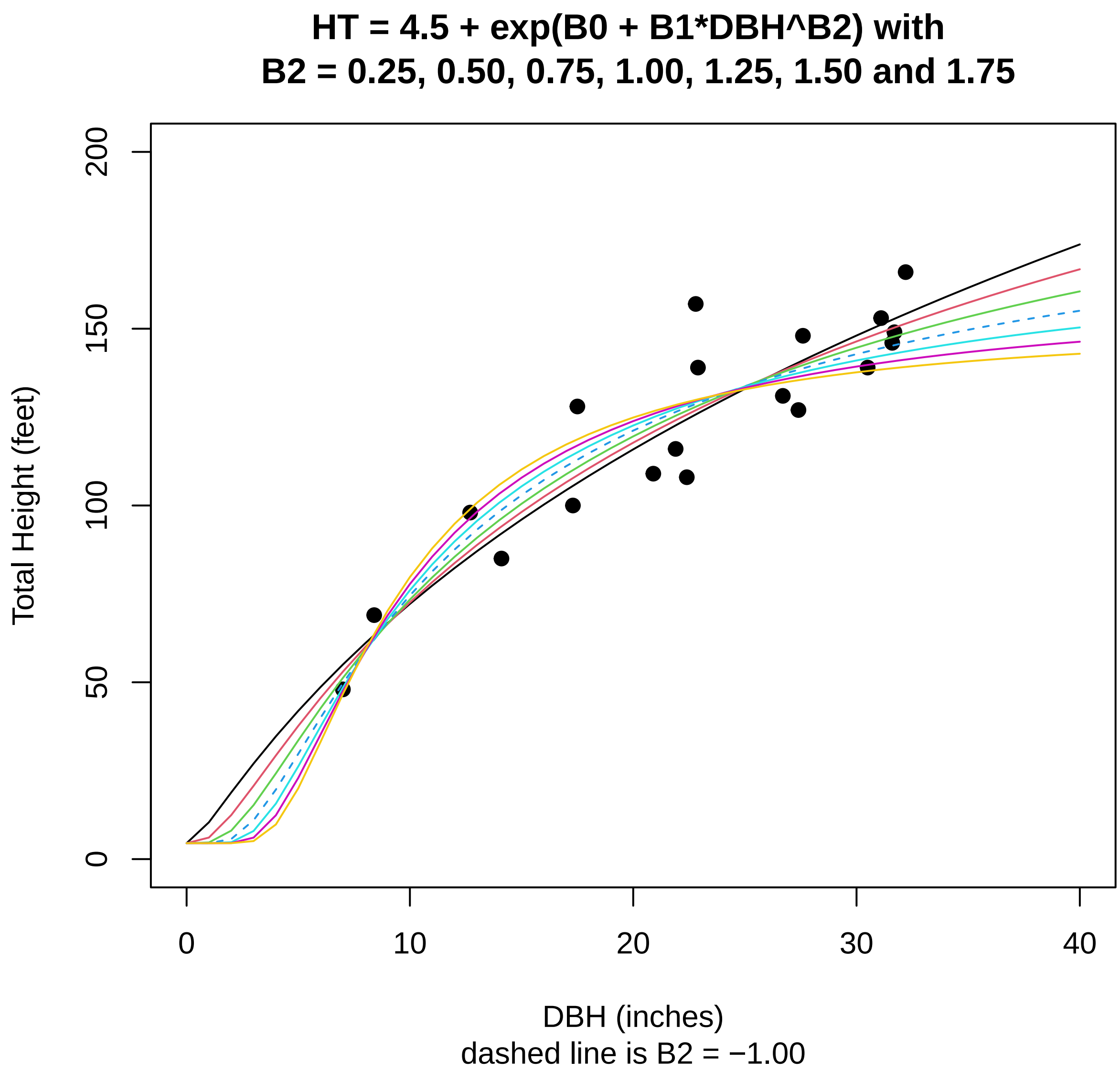


Table 3: Example of test HT-DBH fits.

j	b1	b2	b3	RSE
1	7.346	-5.569	-0.25	0.1151
2	5.958	-5.494	-0.5	0.1118
3	5.497	-7.113	-0.75	0.1111
4	5.27	-10.2	-1	0.1129
5	5.136	-15.39	-1.25	0.1167

j	b1	b2	b3	RSE
6	5.049	-23.86	-1.5	0.1219
7	4.989	-37.59	-1.75	0.1279

step 2: Check that the fitted B1 parameter is reasonable and needs to consider if B1 is positive or B1 becomes a very large negative value. The B0 and B2 parameters seem to be less prone to unreasonable values, especially if B1 is reasonable. However, B2 can become positive (usually if B1 is unreasonable). If B1 is unreasonable B2 is fixed which often takes care of the issue. If B1 > -0.25 (near 0) then B2 is set to -0.25 and if B1 < -75.0 (very negative) then B2 is set to -1.75. This takes care of most issues. In rare cases the B2 parameter > 0 (narrow DBH range with decreasing HTs with increasing DBH) and in this case B1 is set to 5.0 and B2 is set to -0.50 (typical for young stands where this seems to happen). A better solution is needed but this seems to work here. Martin and Flewelling (1998) suggested that the use of a boundary function could be effective for controlling unreasonable parameter estimates (this was not considered here, but worth looking into).

step 3: Fit the nonlinear model with the best fit from steps 1 and 2 (unless B2 is constrained in atep 2) as the starting values if b1 was not > 0.

step 4: Make final checks that the nonlinear fit converged with no errors and that the B1 is reasonable (repeat step 2) and if not use the linear fit.

The following are the results for fitting all plots individually.

These are 121335 total plots fit and the tables below summarize the fit type and the parameter estimates.

The fit types (with or without constraints) are:

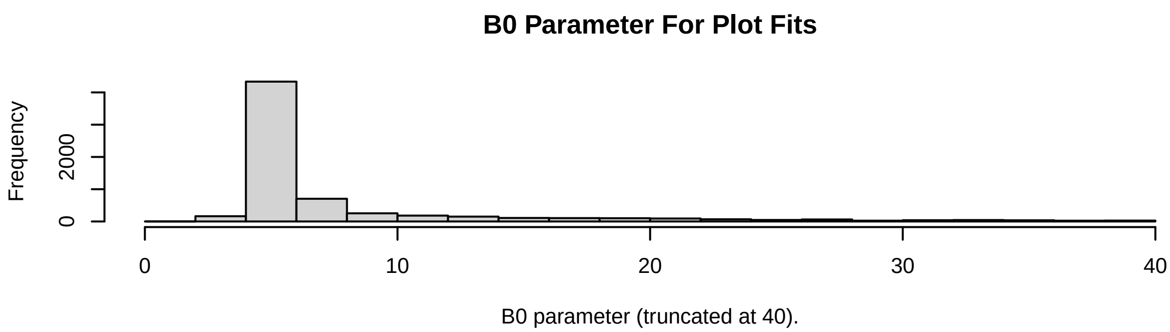
- 0 nonlinear fit
- 1 linear fit
- 2 linear fit with constrained b1 (b1 > B1max) with constant b2 based on B1max
- 3 linear fit with constrained b1 (b1 < B1min)) with constant b2 based on B1min
- 4 linear fit with constrained b1 (b1 < B1min)) with constant b2 based on CC value
- 5 linear fit with constrained b1 (b0 <= 0) with constant b0 and b2 (rare?)

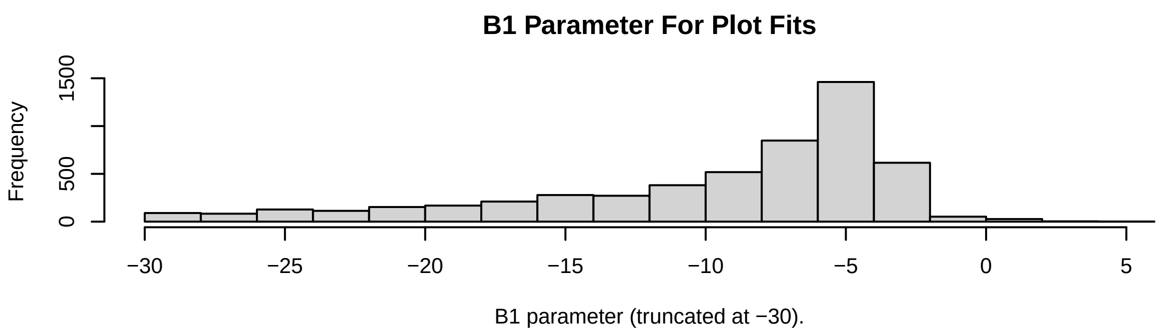
Table 4: Number of fitted plots by fit type as defined above.

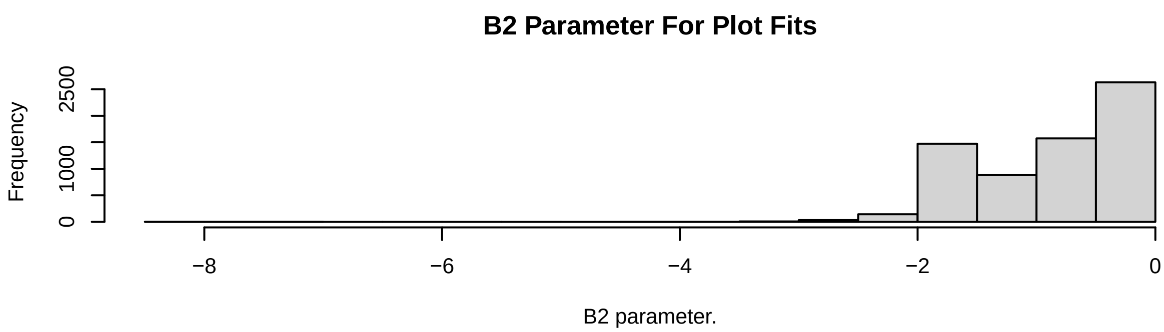
0	2	3	5
5650	39	1055	1

Table 5: Summary of parameters for individual plot fits.

Param	Min.	X1st.Qu.	Median	Mean	X3rd.Qu.	Max.
B0	1.668	4.902	5.398	9.039	7.327	77.54
B1	-768.3	-23.3	-9.307	-23.12	-5.368	4.112
B2	-8.085	-1.481	-0.7164	-0.8439	-0.219	-0.00649



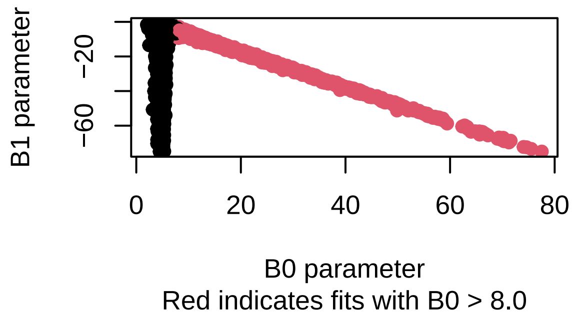


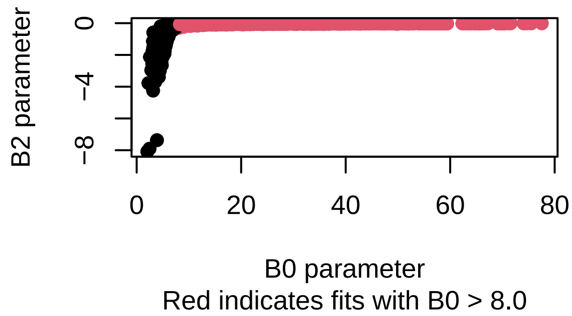


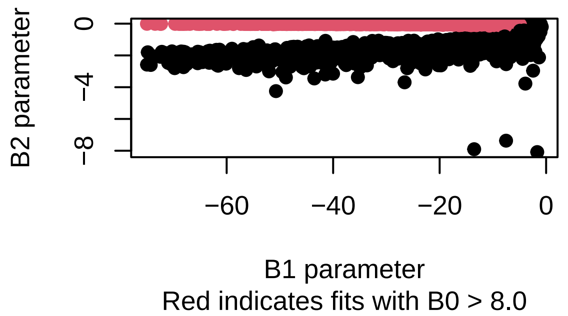
The distribution of DBH and HT pairs for fitting can impact the fits but the parameters are also correlated with this model form. These are for the plots that were fit with nonlinear regression (fit type = 0) and were not constrained.

Table 6: Correlation matrix of parameters for individual plot fits (successful nonlinear fits only).

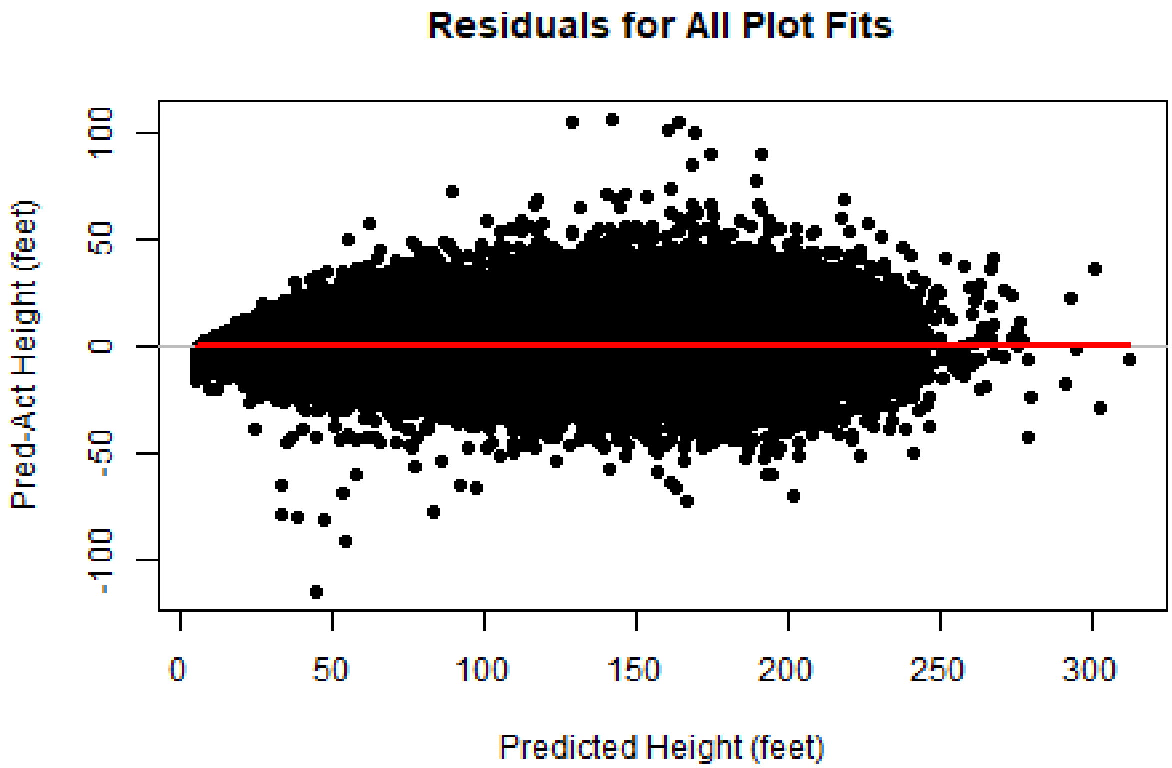
	b0	b1	b2
b0	1	-0.5585	0.4926
b1	-0.5585	1	0.2646
b2	0.4926	0.2646	1



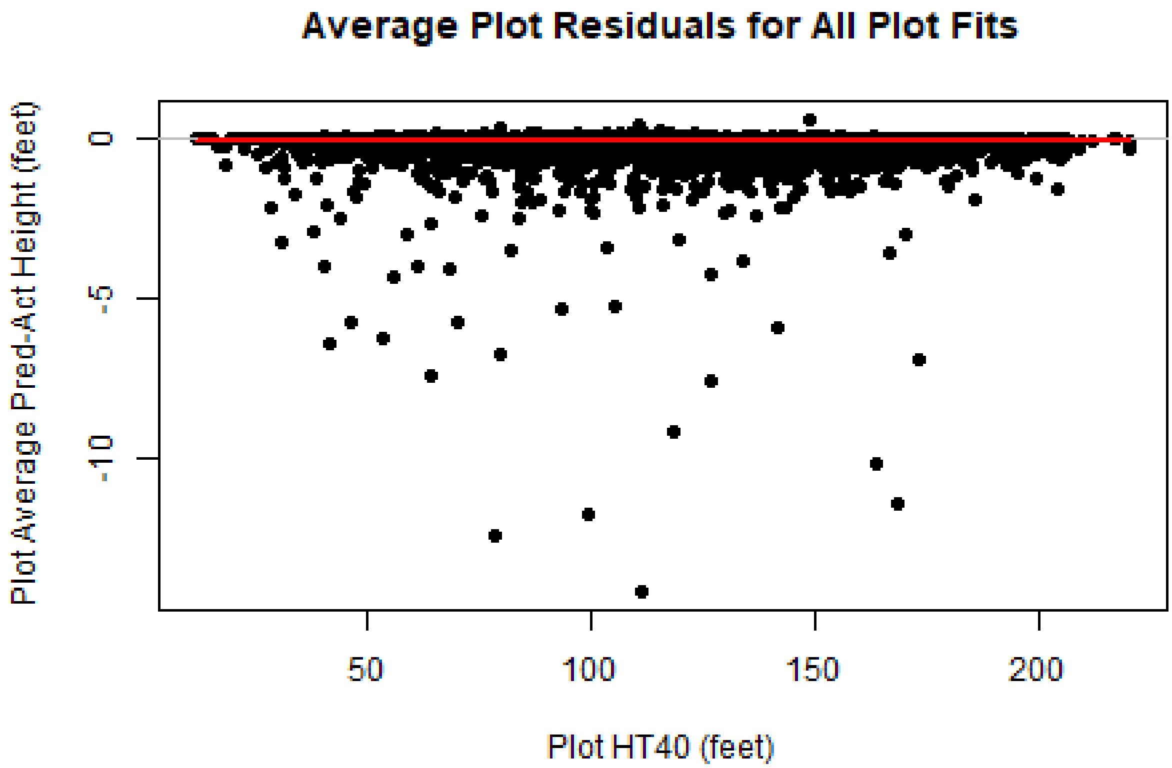




A plot of all residual combined is shown below compared to the predicted tree heights, The second graph shows the plot average residual (we would expect these to be about 0) plotted against the plot HT40 which representing stand development (age may be missing or unreliable for older or uneven-aged/mixed species plots). The total (for all plots) RSE is 126.5735574 feet.



Residuals for All Plot Fits
Pred-Act Height (feet)
100
50
0
-50
-100
0
50
100
150
200
250
300
Predicted Height (feet)



3.2 Regional Model Fit

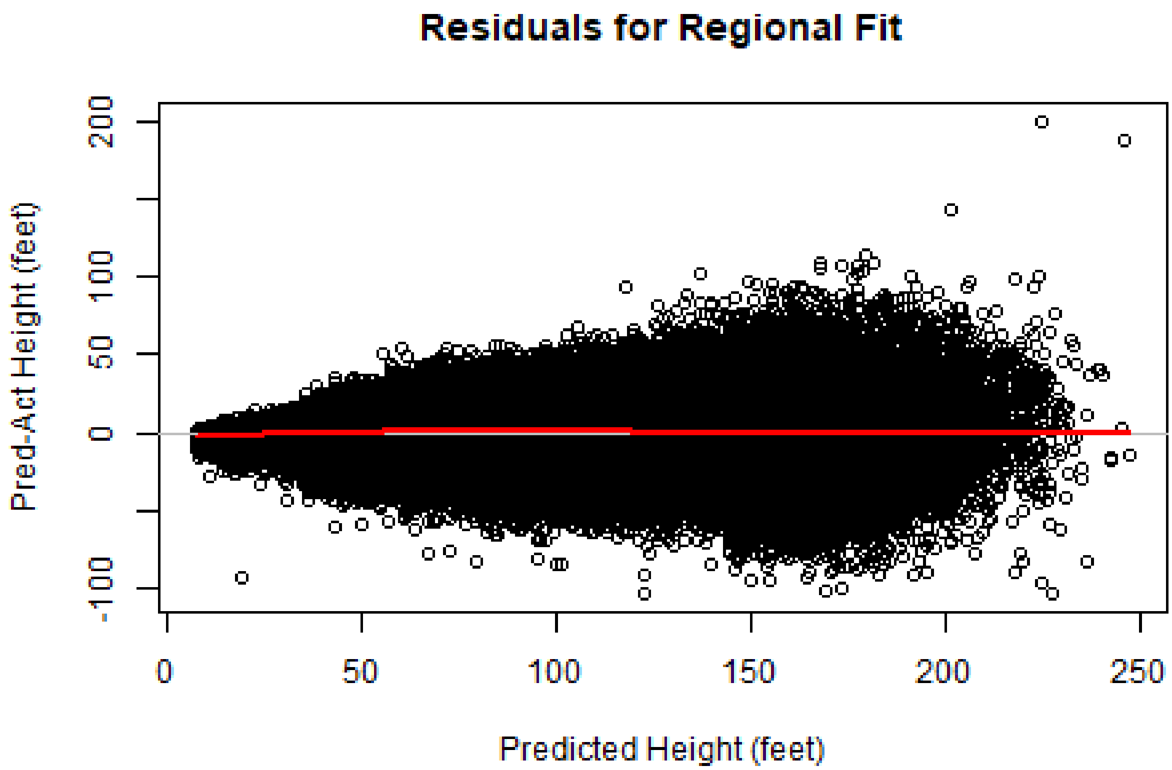
In this section all data will be combined and used to fit a traditional regional HT-DBH model using nonlinear least squares. Starting values were obtained by transforming the model to the linear form by setting B2 = -1.0.

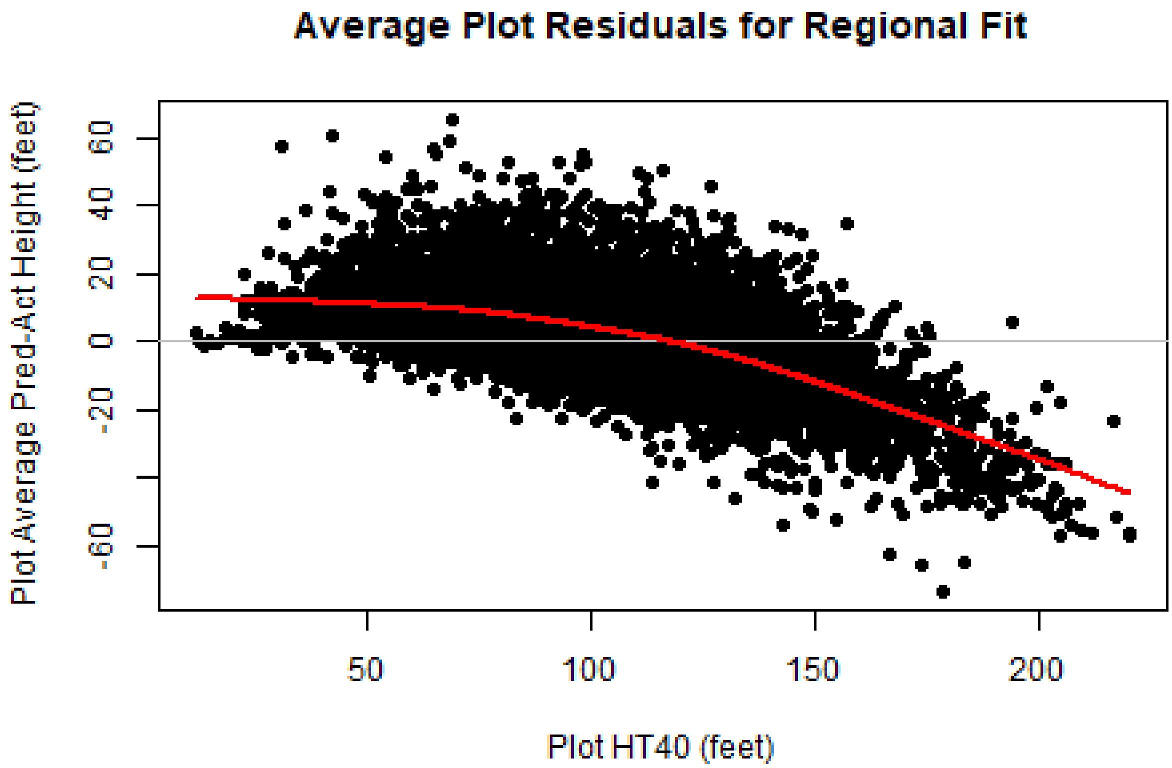
$ log(HT-4.5) = B_0 + B_1*DBH^{-1.0} $.

```
##
## Formula: HT ~ 4.5 + exp(A + B * DIA^C)
##
## Parameters:
##    Estimate Std. Error t value             Pr(>|t|)
## A  6.551591   0.021441  305.57 <0.0000000000000002 ***
## B -5.611459   0.009219 -608.71 <0.0000000000000002 ***
## C -0.366687   0.004173  -87.88 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.57 on 121332 degrees of freedom
##
## Number of iterations to convergence: 12
## Achieved convergence tolerance: 0.0000000149

## AIC =  1065989
```

Table 7: Regional model fit summary.

Fit	N	meanBIAS	absBIAS	RMSE
Regional	121335	-0.0744	14.7	19.57





This fit is as expected (signs) with parameters similar to those in Table 1. The graph above on the left shows no major trends or bias. However, the graph on the left plots the average PLOT residual against the plot average height of the 40-largest DBH trees (HT40) which gives an idea of the stage of stand development (similar to age, but age is not always available for uneven-aged stands and is unreliable for older stands). The trend with HT40 demonstrates the problem is applying the regional curves to individual stands that have different trends than the average.

3.3 Calibrate Regional HT-DBH Curves

One way to deal with the problem of regional HT-DBH curves not fitting individual stands this is to calibrate the regional fit using measured heights from the subject stand. Temsegen and others (2008) use one such calibration method that is computed as:

$$K = sum((pHT - 4.5) * (HT - 4.5))/sum((pHT - 4.5)^2)$$

where K is the height calibration, pHT is the predicted height, and HT is a measured height. The calibration is applied as

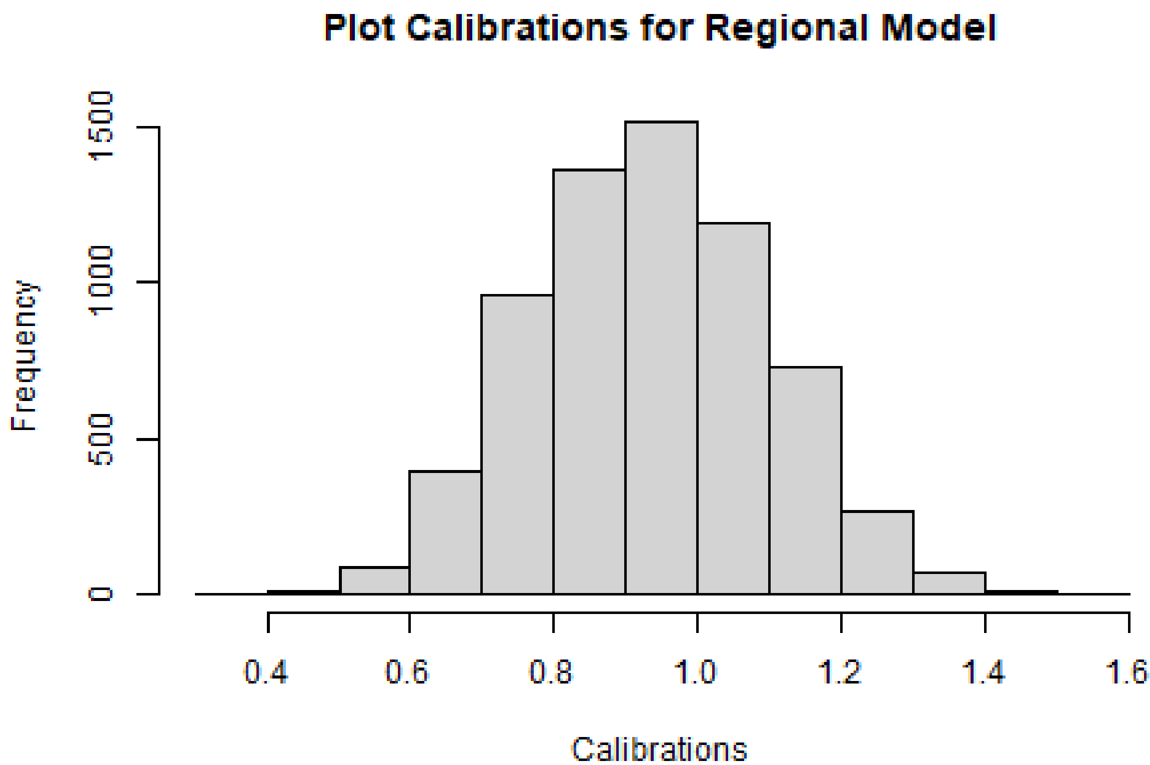
$$adjHT = 4.5 + K * (pHT - 4.5)$$

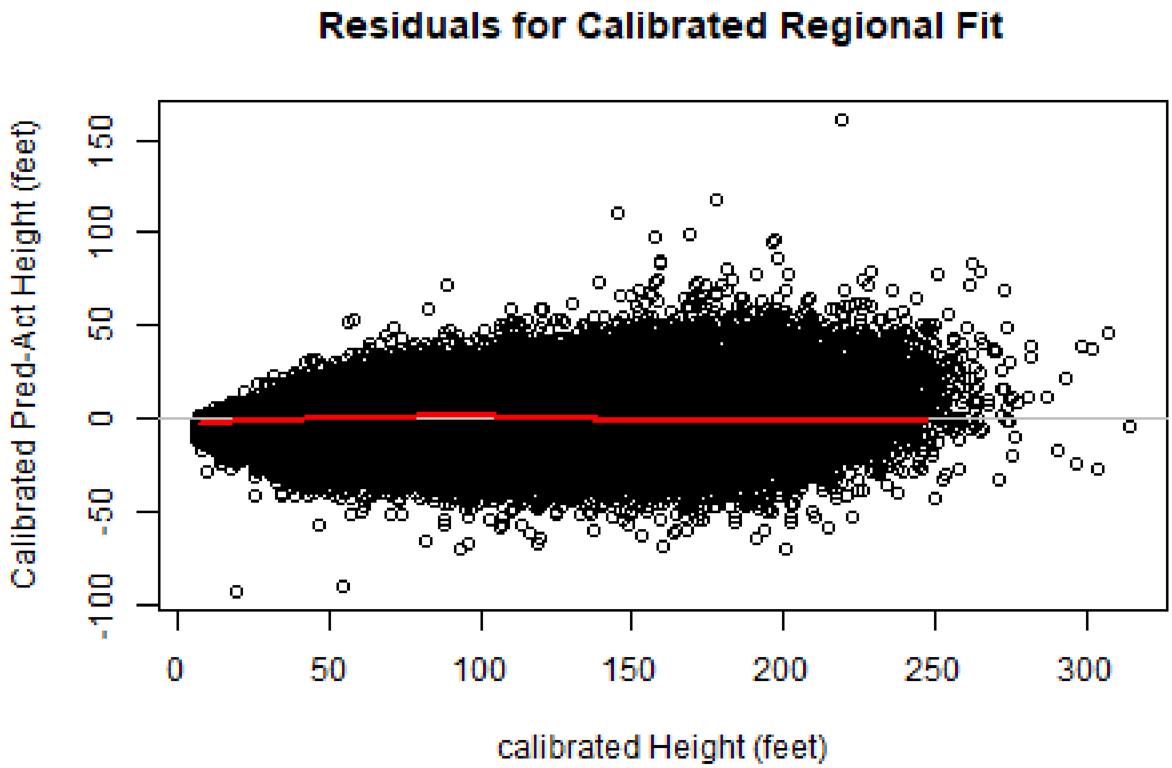
where adjHT is the calibrated predicted height.

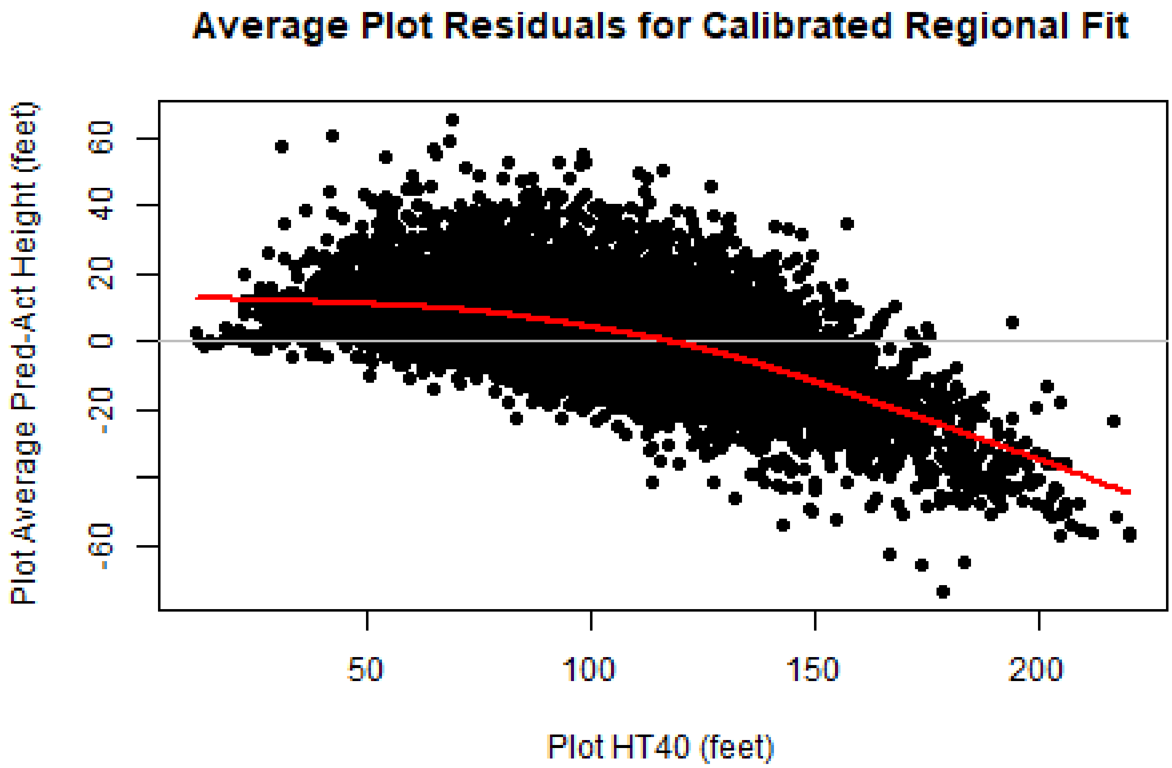
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.3555  0.8153  0.9311  0.9335  1.0506  1.5336
```

Table 8: Summary of Plot Regional Model Calibrations.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3555	0.8153	0.9311	0.9335	1.051	1.534




Residuals for Calibrated Regional Fit
Calibrated Pred-Act Height (feet)
-100
-50
0
50
100
150
0
50
100
150
200
250
300
calibrated Height (feet)




```
## Error: object 'summary_CalibrationFit' not found
```

Error: ! object 'summary_CalibrationFit' not found Backtrace: x 1. -pander::pander(...) The calibration takes out most of the trends in the residual. The analysis of Temesgen and others (2008) suggests about 8 trees are needed for calibrations in their data.

3.4 Enhanced Regional Model Fit

Temesgen and others (2007) compared regional HT-DBH curves fit with DBH only and with DBH enhanced with stand density (BA and CCF) and tree social position (BAL and CCFL) predictor variables and found the enhanced equations increased accuracy of predictions over the DBH only model. They modified the model used for this analysis as follows:

$$HT = bh + (b_0 + b_1 * position + b_2 * density) * exp(B_1 * DBH^{B_2})$$

where position is BAL or CCFL and density is BAPA or CCF.

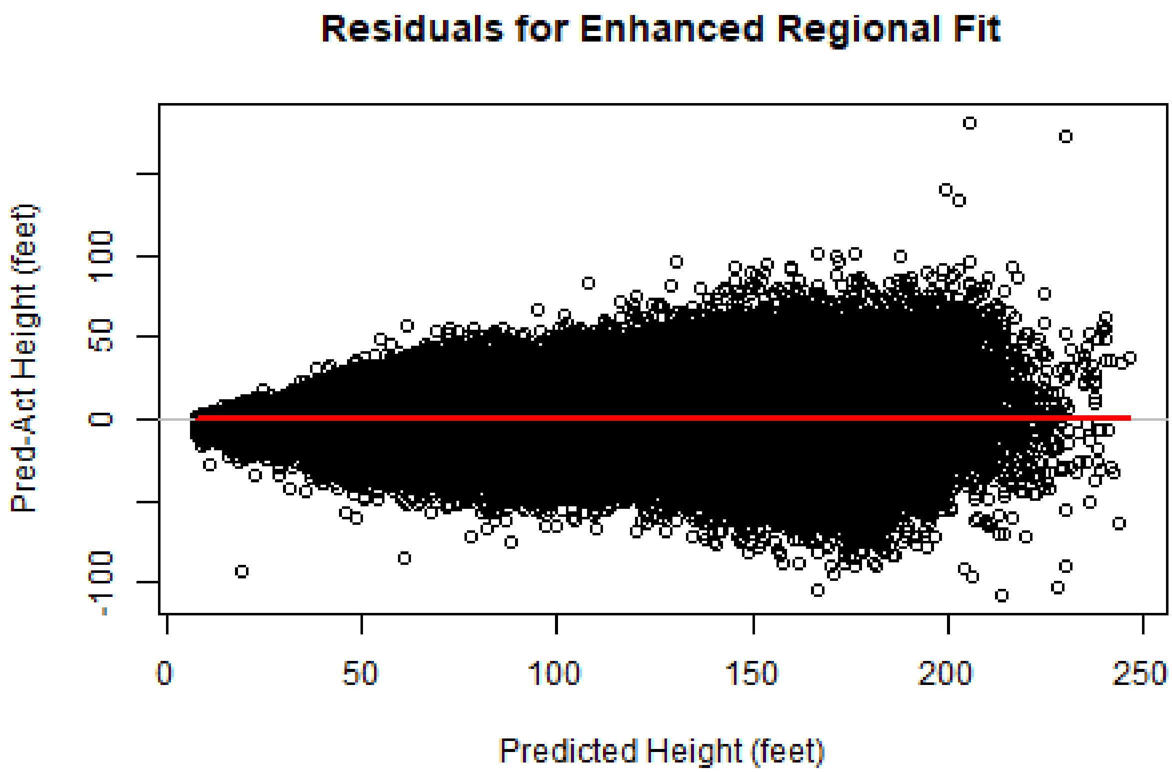
```
##
## Formula: HT ~ 4.5 + (a0 + a1 * BAL + a2 * BAPA) * exp(B * DIA^C)
##
## Parameters:
##      Estimate Std. Error t value            Pr(>|t|)
## a0 741.167578  18.957343   39.10 <0.0000000000000002 ***
## a1   0.674803   0.020502   32.91 <0.0000000000000002 ***
```

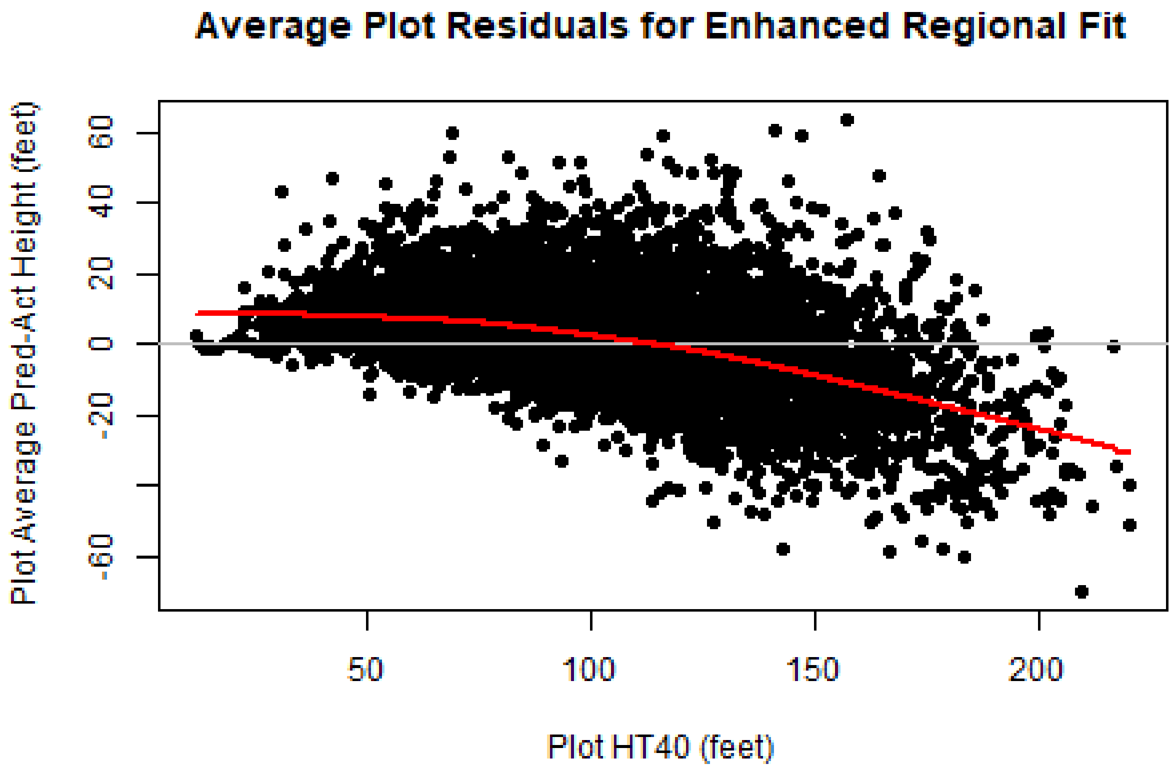
```
## a2   0.126915   0.005595   22.68 <0.0000000000000002 ***
## B   -5.637999   0.010091 -558.72 <0.0000000000000002 ***
## C   -0.336949   0.003975  -84.78 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.04 on 121330 degrees of freedom
##
## Number of iterations to convergence: 31
## Achieved convergence tolerance: 0.0000000149

## AIC =  1046267
```

This fit shows an improvement in RSE and AIC over the un-enhanced regional model.

Table 9: Regional enhanced model fit summary.

Fit	N	meanBIAS	absBIAS	RMSE
Enhanced	121335	-0.007955	13.41	18.04





3.5 Fit with a Mixed Effects Model

Another way that has been used to calibrate regional fits to local conditions is to use mixed-effects models to account for the locations where plots or stands are random effects. Lappi (1991) gives an example of how the random effects can be recovered for new locations (also see Lynch and others (2005) and Temesgen and others (2008)).

The Schumacher type model is challenging to fit as a mixed model and after trying a number of starting values (nonlinear least squares, fixed parameters, single nlme parameter fits) the function would not converge with random effects on two parameters (have not tried nlmer in the lme4 package which uses a different algorithm). Therefore it was fit with a single random effect of the B) (asympote) parameter.

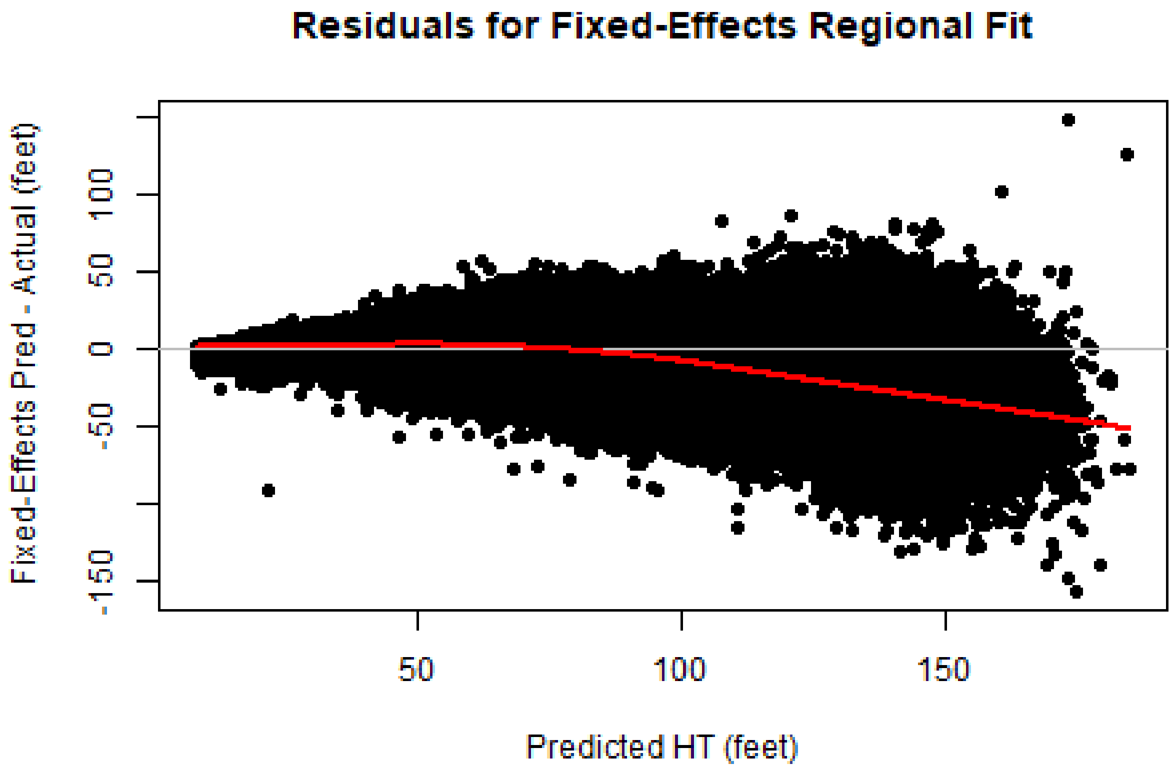
```
##
## **Iteration 1
## LME step: Loglik: -479479.5, nlminb iterations: 1
## reStruct  parameters:
##   PLOTID
## 4.011841
##  Beginning PNLS step: ..  completed fit_nlme() step.
## PNLS step: RSS =  15822493
##  fixed effects: 5.789925  -4.577711  -0.4491481
##  iterations: 5
## Convergence crit. (must all become <= tolerance = 1e-05):
##       fixed    reStruct
## 0.004868925 0.003075150
```

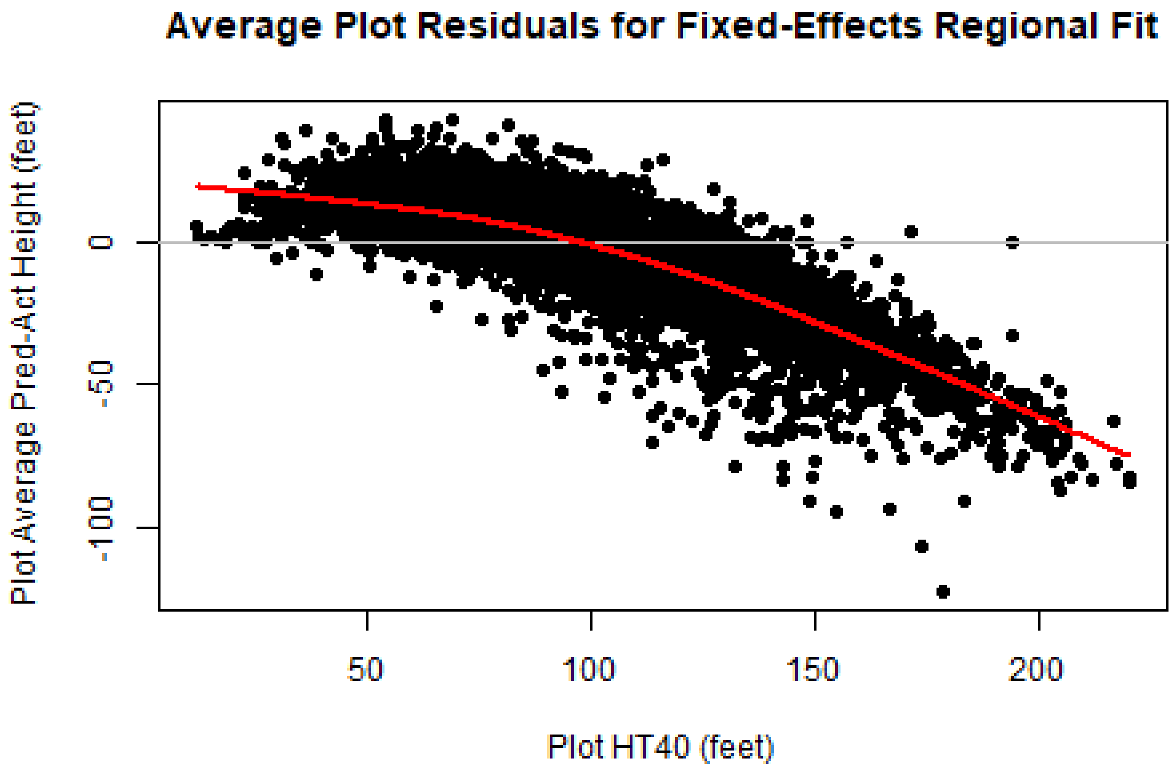
```
##
## **Iteration 2
## LME step: Loglik: -479485.5, nlminb iterations: 1
## reStruct  parameters:
##   PLOTID
## 4.024216
##  Beginning PNLS step: ..  completed fit_nlme() step.
## PNLS step: RSS =  15822493
##  fixed effects: 5.789925  -4.577711  -0.4491481
##  iterations: 1
## Convergence crit. (must all become <= tolerance = 1e-05):
##    fixed reStruct
##        0        0
```

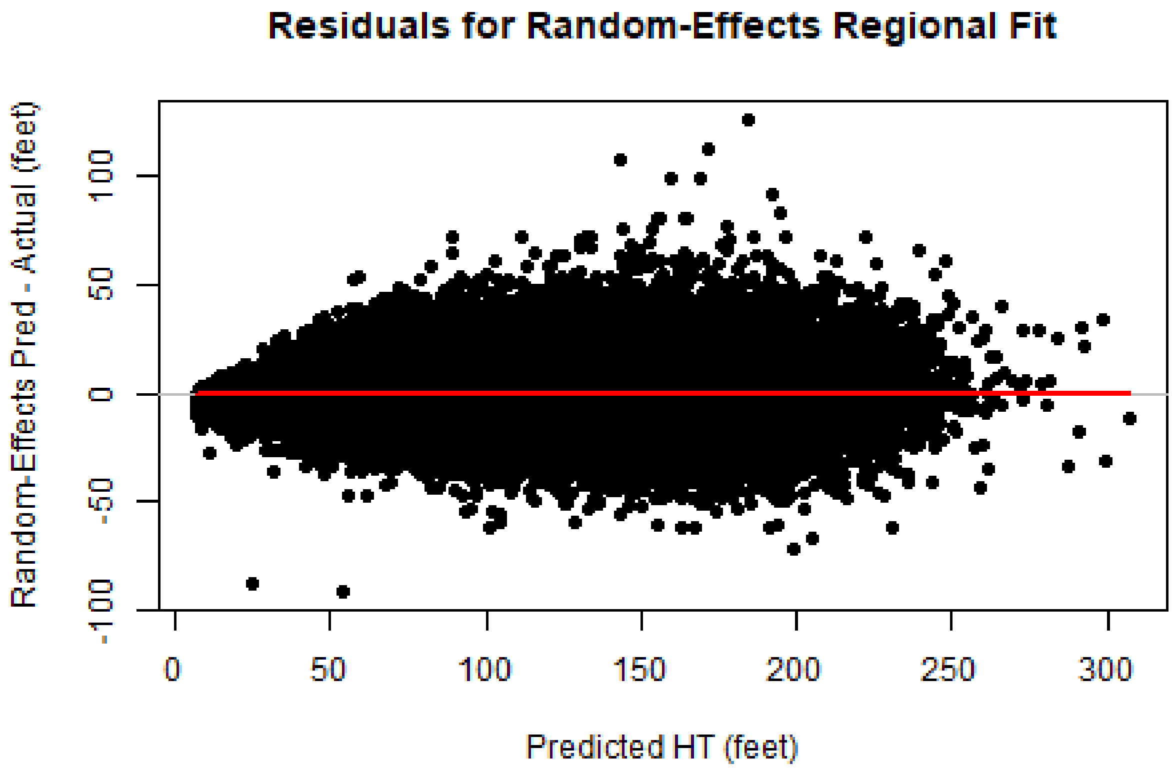
```
## Nonlinear mixed-effects model fit by maximum likelihood
##   Model: HT ~ 4.5 + exp(A + B * DBH^C)
##   Data: dataX
##      AIC      BIC    logLik
##   958981 959029.5 -479485.5
##
## Random effects:
##  Formula: A ~ 1 | PLOTID
##                 A Residual
## StdDev: 0.2041502 11.41943
##
## Fixed effects:  A + B + C ~ 1
##       Value   Std.Error     DF   t-value p-value
## A  5.789925 0.010917184 114738  530.3496       0
## B -4.577711 0.011839022 114738 -386.6630       0
## C -0.449148 0.003666803 114738 -122.4904       0
##  Correlation:
##   A     B
## B 0.582
## C 0.951 0.751
##
## Standardized Within-Group Residuals:
##          Min           Q1          Med           Q3          Max
## -11.07344548  -0.52479481   0.01442457   0.54877799   8.00008294
##
## Number of Observations: 121335
## Number of Groups: 6595
```

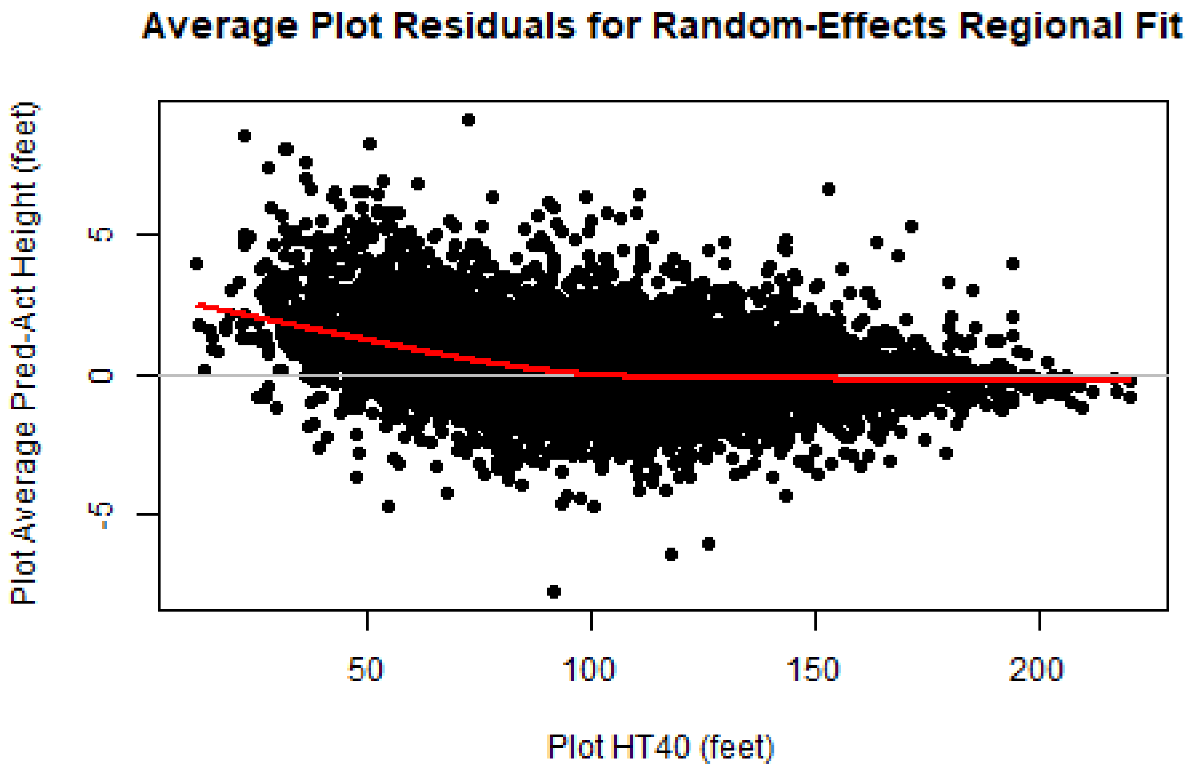
```
##        Value   Std.Error     DF   t-value p-value
## A  5.7899246 0.010917184 114738  530.3496       0
## B -4.5777115 0.011839022 114738 -386.6630       0
## C -0.4491481 0.003666803 114738 -122.4904       0
```

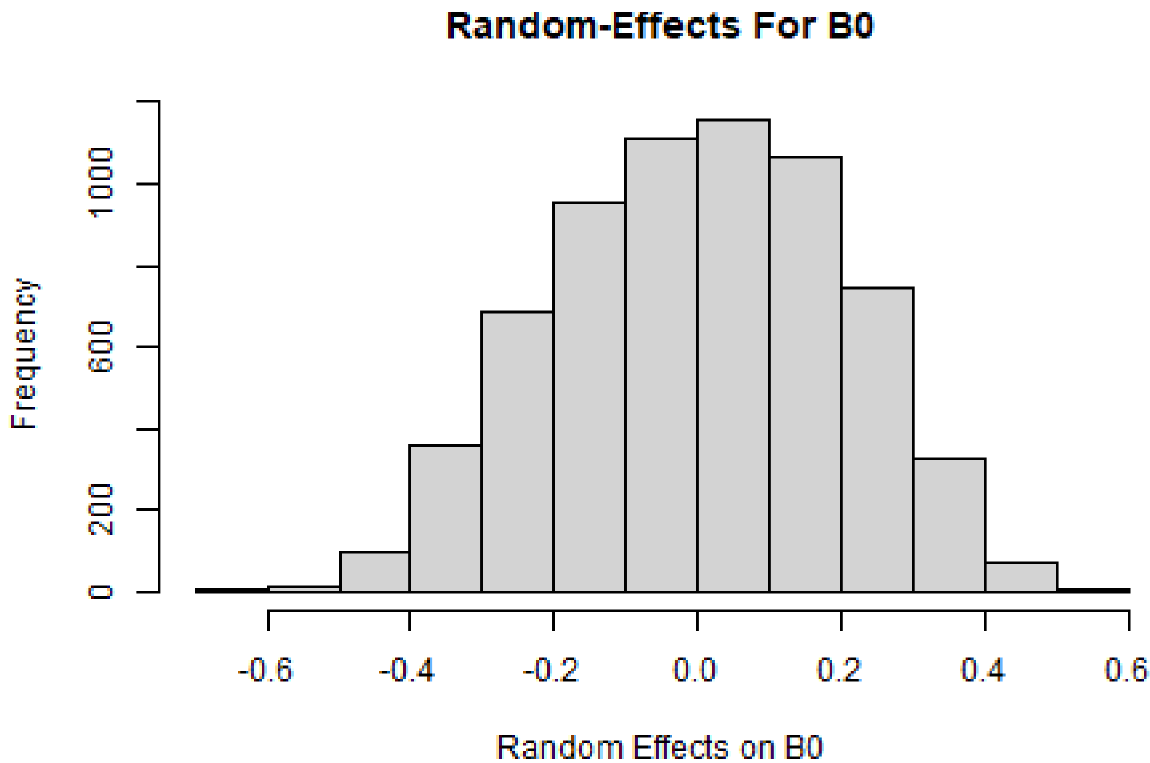
```
## AIC =  958981
```


Residuals for Fixed-Effects Regional Fit
Fixed-Effects Pred - Actual (feet)
100
50
0
-50
-150
50
100
150
Predicted HT (feet)


Average Plot Residuals for Fixed-Effects Regional Fit
Plot Average Pred-Act Height (feet)
0
-50
-100
50
100
150
200
Plot HT40 (feet)


Residuals for Random-Effects Regional Fit
Random-Effects Pred - Actual (feet)
-100
-50
0
50
100
0
50
100
150
200
250
300
Predicted HT (feet)


Average Plot Residuals for Random-Effects Regional Fit
Plot Average Pred-Act Height (feet)
5
0
-5
50
100
150
200
Plot HT40 (feet)


Random-Effects For B0
Frequency
0
200
600
1000
-0.6
-0.4
-0.2
0.0
0.2
0.4
0.6
Random Effects on B0

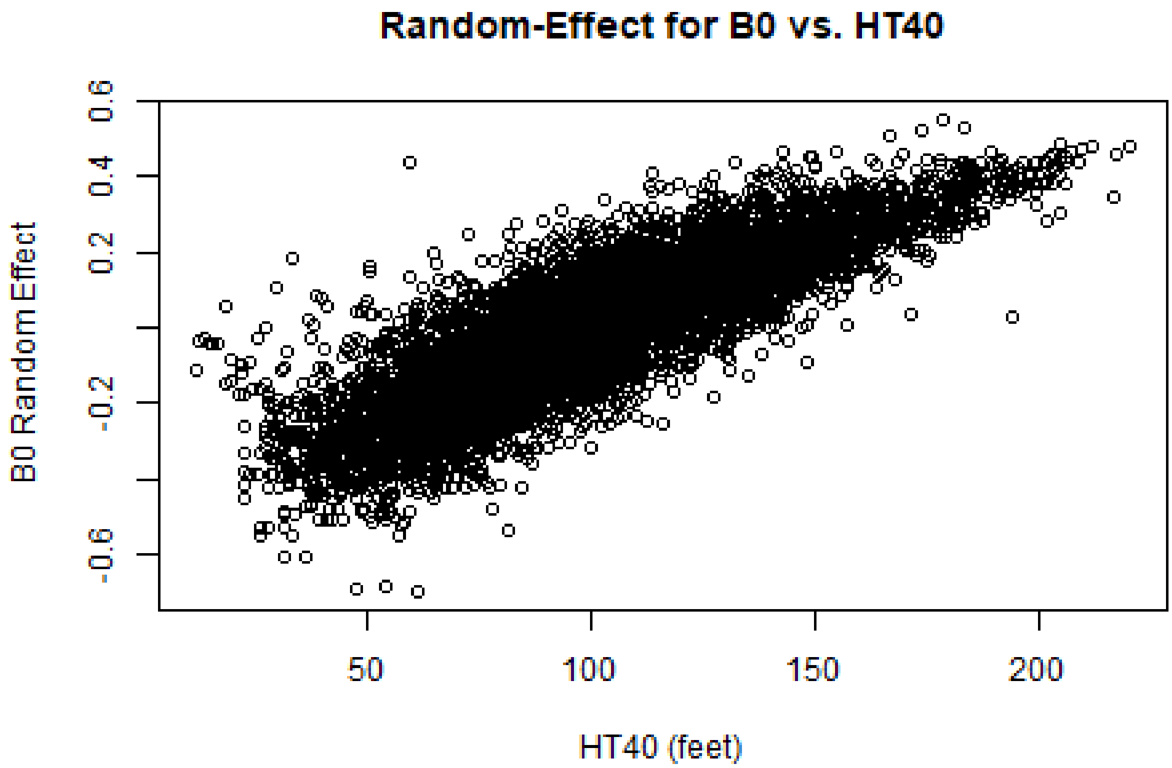


Table 10: Regional mixed-effects model fit (random effects) summary.

Fit	N	meanBIAS	absBIAS	RMSE
Random-Effects	121335	0.0535	8.173	11.12

The fixed effects model behave like the regional model as we might expect. Using the random effects removes much of the bias across HT40.

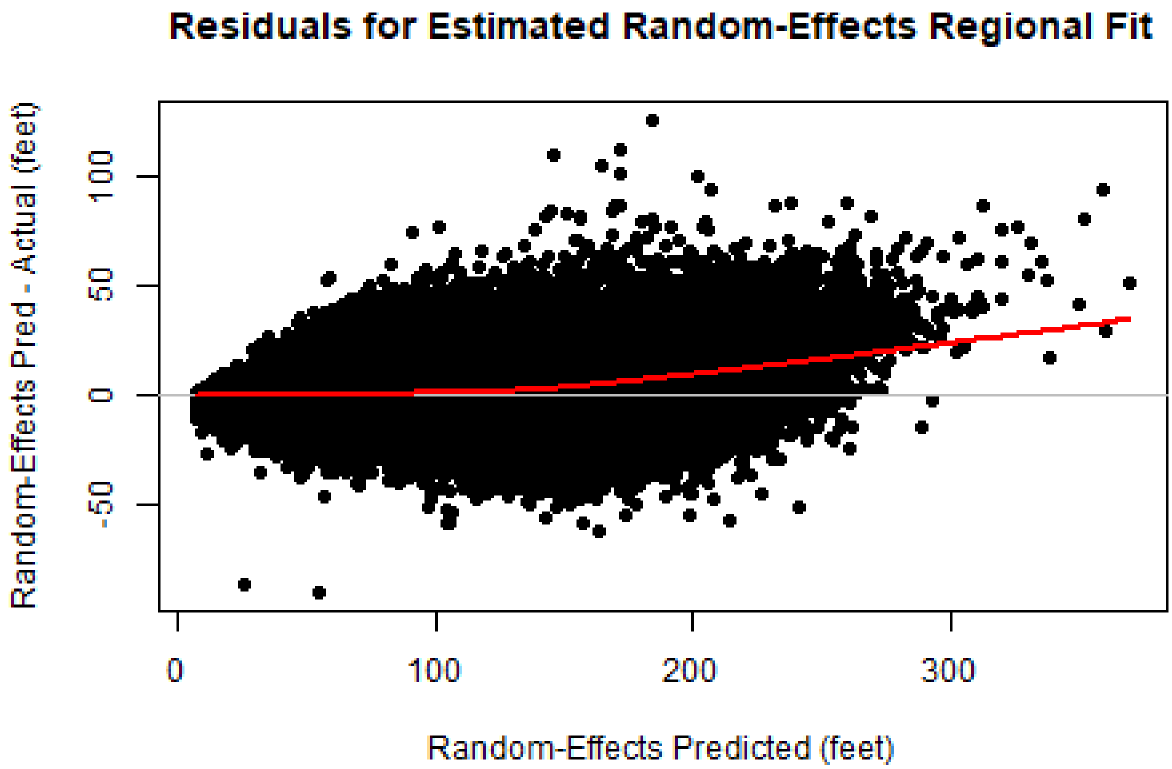
3.6 Estimate Random Effects (BLUP) For PLot

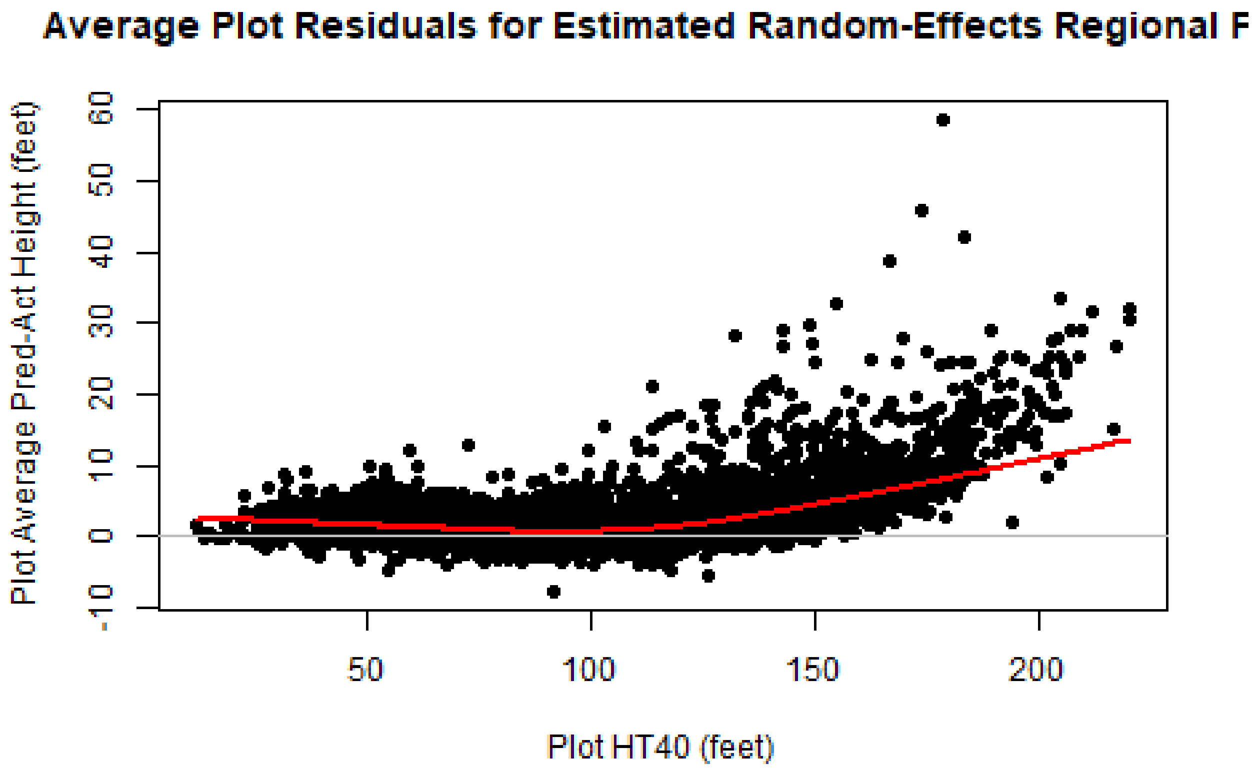
In the above the random effects are based on the fit. However, this does not work for applying the HT-DBH predictions to new stands. For "new" stands with a sample of DBH and HT measurements the random effects (BLUPs) can be computed as demonstrated by Lappi (1991) (also see Lynch and other 2005; Temsegen and other 2008). This was done by an R function that implements the above methods.

Here the random-effects are recovered for each stand using all trees (to look at sample size see Temesgen and others 2008) and comparing the predicted "localized" heights to the actual measuremented heights.

Table 11: Regional mixed-effects model fit with random effects computed by plot summary.

Fit	N	meanBIAS	absBIAS	RMSE
Random-Effects_Recovered	121335	2.111	8.539	11.85

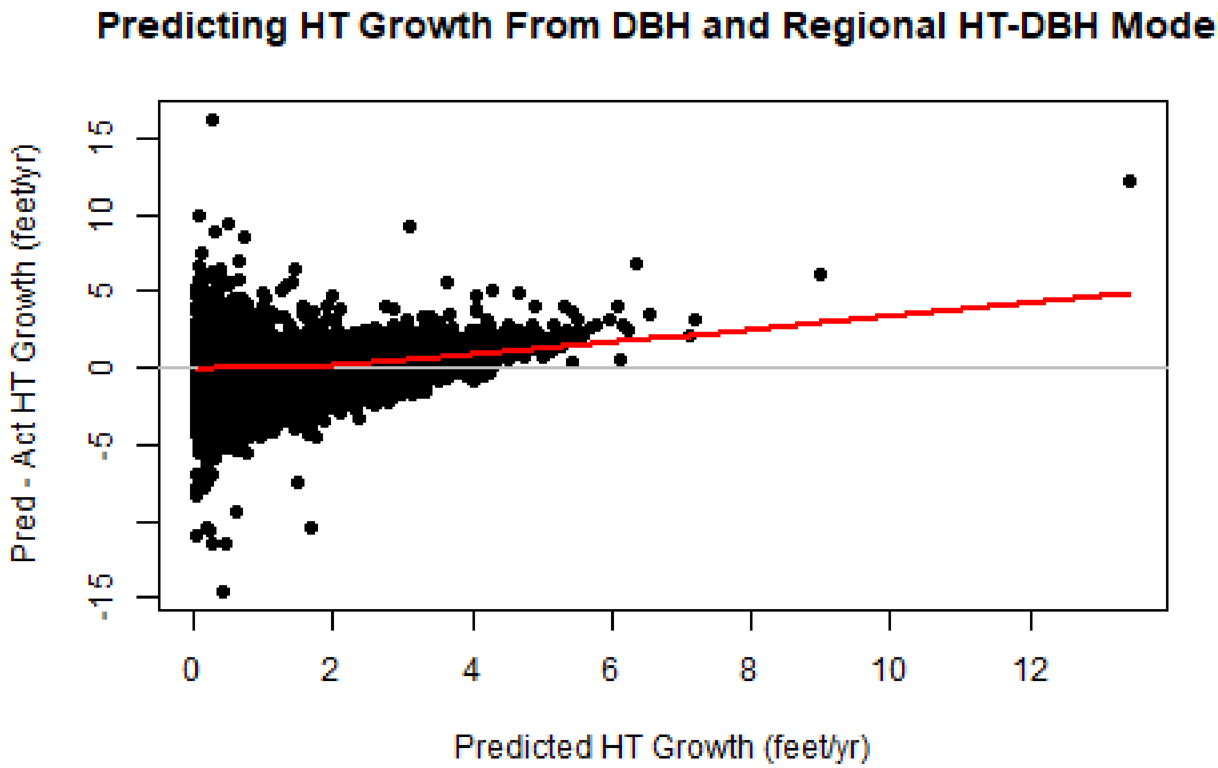




3.6 Height Growth with Regional Estimates

Estimating missing heights using a regional HT-DBH model is a bad idea without some type of "localization". However, if there were good initial heights (measured or appropriately predicted) does it matter for getting height growth? The question therefore is the relative change in heights computed from the DBH at the start and the end of the period OK as long as the initial height is correct)? This is used in some tree-list type models, especially for minor species (e.g. ORGANON hardwoods).

To investigate this FIA data for OREGON Douglas-fir trees that are alive and have a measured DBH and HT at the start and the end of the growth period (and the growth period is at least one year). The HT at the start and the end of the period are predicted from the DBH using the regional HT-DBH curve for Pacific Northwest Douglas-fir from Hanus and others (1999). The annualized predicted and actual height growth is compared.



This suggests that use of regional HT-DBH curve can be used to estimate height grow with minimal bias and future height if the initial height is a good estimate.

4.0 Discussion

Both the calibration and the mixed effects models appear to remove most of the "bias" for individual plots/stands. For this data set fitting the mixed effects model was a bit more challenging to get convergence and recoving the BLUPs is computationally more involved than calibrations. For this study all trees were used to "localize" but in reality this would be done with a sample. Temsegen and others (2008) compared these methods of localization for sample size. They found that the random-effects required fewer trees than the calibration method (to reach a similar RMSE), however the differences were small (4-6 for random-effects and 8-10 for calibration).

The Schumacher type model is very useful for fitting HT-DBH curves for stands because of its flexibility, easy to linearize to use or get starting values and a value of -1.0 often works fine worn there is little data. However, it is a challenge to fit with random effects on more than one parameter, at least for this data set. It is not clear if it is the data or the model form (e.g. correlation of the parameters). One wonders if this was the reason Temesgen switch from the positive ranking of the Schumacher model type (Temsegen and Gadow 2004; Temesge and others 2007) to the Chapman-Richards model in Temesgen and others (2008).

Both the random-effects and calibration methods require "new" data. Others have suggested other options. For example, Krumland and Wensel (1988) anchored the HT-DBH curve to a stands HT40 and DBH40 based on the stand age and site index (see Hanus and others 1999). This seems to work well for even-aged stands but may be problematic with uneven-aged stands or minor (tolerant) species and requires a site height curve that fits the stands of interest well (also see Hilt and Dale 1982).

This analysis only considered a single (unweighted) model form. Visual inspection of the residual graphs suggest weighting would be desirable (likely true with other model forms). It is possible that other model forms might behave differently in predictions (Appendix suggests they are similar as would be expected from other published comparisons previously discussed) or calibrations or mixed effects.

Table 12: Fit comparisons.

Fit	N	meanBIAS	absBIAS	RMSE
All Plots	121335	-0.1166	6.751	9.465
Regional	121335	-0.0744	14.7	19.57
Calibrated	121335	-1.384	9.002	12.1
Enhanced	121335	-0.007955	13.41	18.04
Random-Effects	121335	0.0535	8.173	11.12
Random-Effects_Recovered	121335	2.111	8.539	11.85

5.0 References

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5.0 Appendix - Regional Models Compared

The purpose of this report is not to evaluate model forms for HT-DBH models. However, it can be insightful to briefly consider what the expected differences might be for different model forms. There have been many HT-DBH model forms proposed and evaluated over the years (e.g., Curtis 1967; Huang and Titus (1992), Zhang 1997, Martin and Flewelling 1998).

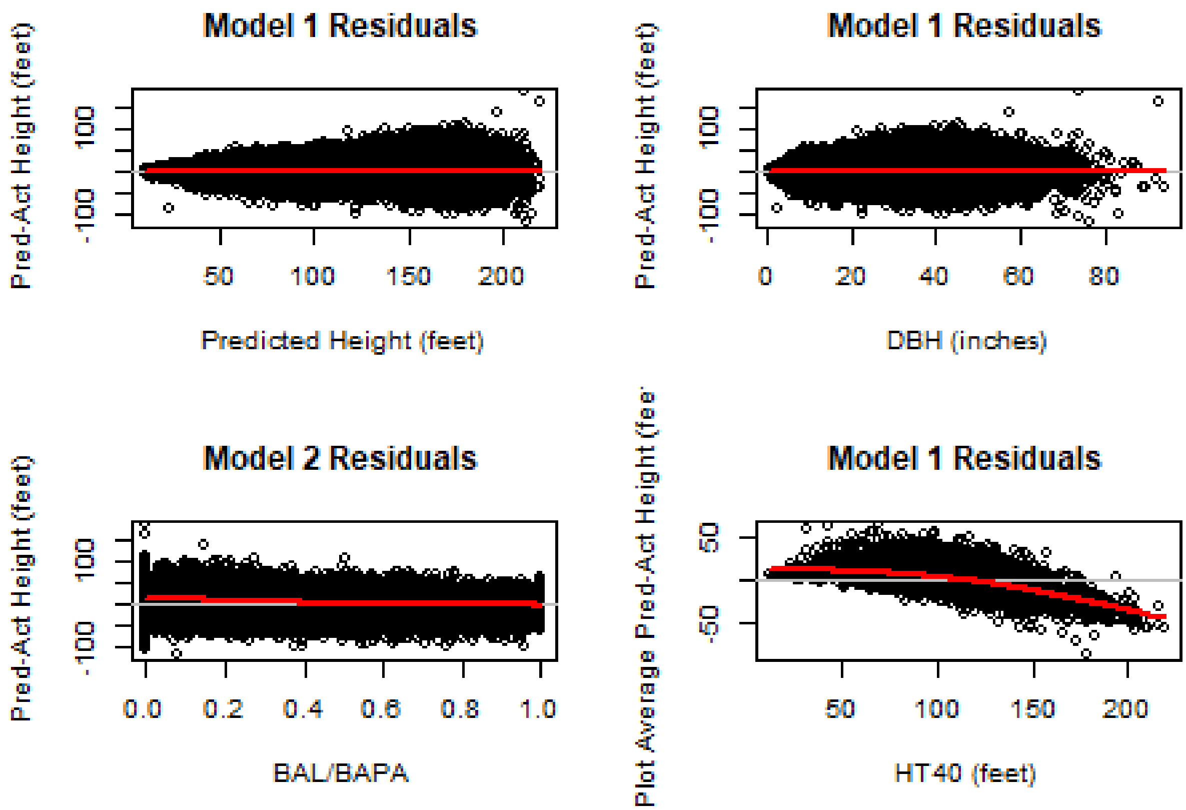
Temesgen and others (2007) evaluated four widely used nonlinear HT-DBH models for 10 species using regional data from southwest Oregon. Below is a comparisons of these four models fit to the FIA Douglas-fir data for Oregon where HT is the total height (feet), DBH is the diameter at breast height (inches), bh is the breast height (4.5 feet) and B0, B1 and B2 are the fitted parameters. All of the models are constrained to go to 4.5 feet at DBH = 0, are sigmoidal and asymptotic. The fits are not weighted.

Model 1: Weibull Model Form

$$HT = bh + B_0 * (1.0 - exp(B_1 * DBH^{B_2}))$$

Table 13: HT-DBH Model 1 fit summary.

Model	A	B	C	RegSE	AIC
1	225.4	-0.03055	1.021	19.52	1065477

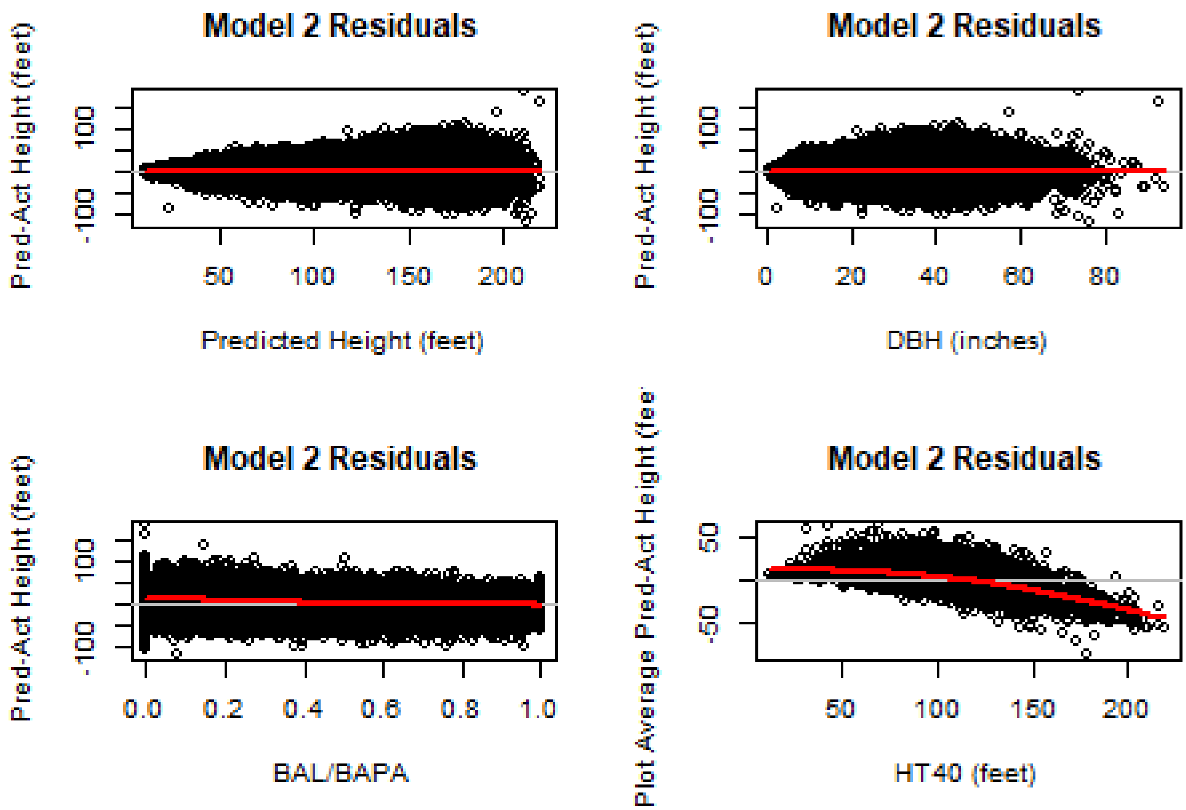


Model 2: Chapman-Richards Model Form

$$HT = bh + B_0 * (1.0 - exp(B_1 * DBH))^{B_2}$$

Table 14: HT-DBH Model 2 fit summary.

Model	A	B	C	RegSE	AIC
2	226.3	-0.03333	1.029	19.52	1065475

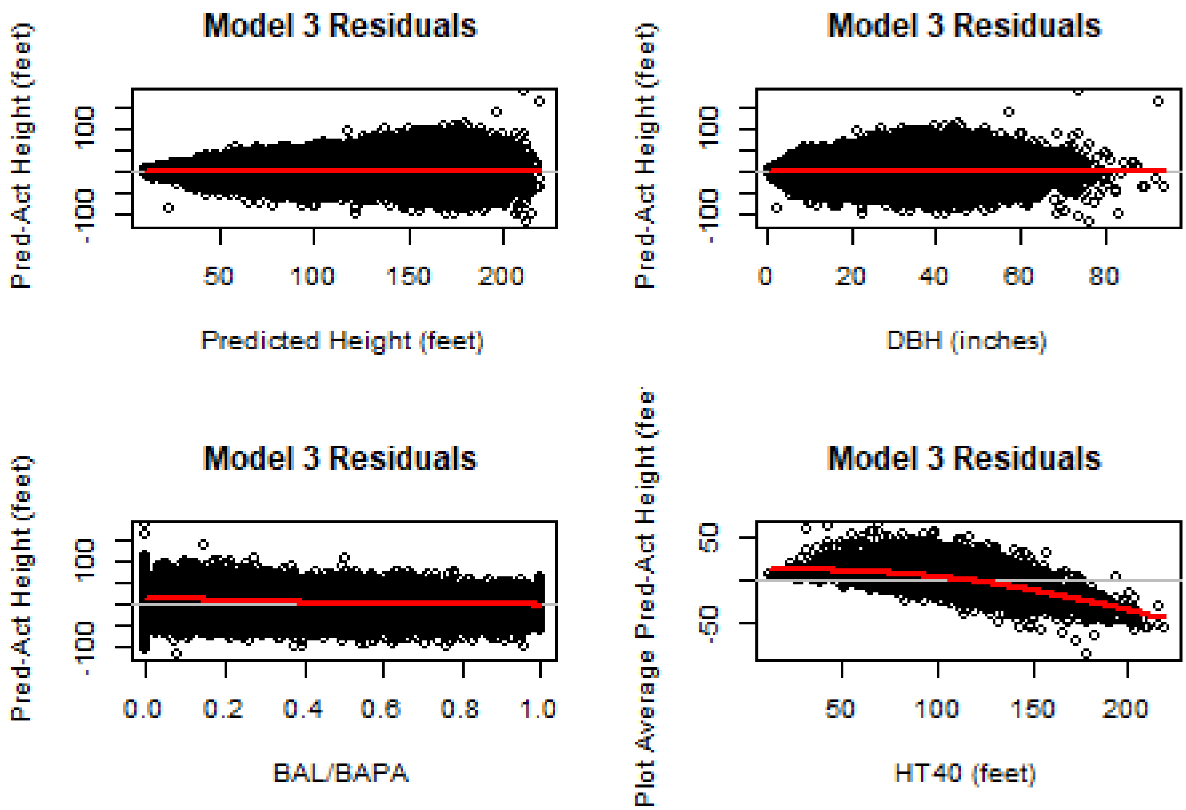


Model 3: Exponential Model Form (see Wykoff 1986, page 12)

$$HT = bh + exp(B_0 + B_1/(DBH + B_2)$$

Table 15: HT-DBH Model 3 fit summary.

Model	A	B	C	RegSE	AIC
3	5.671	-26.78	7.256	19.55	1065815

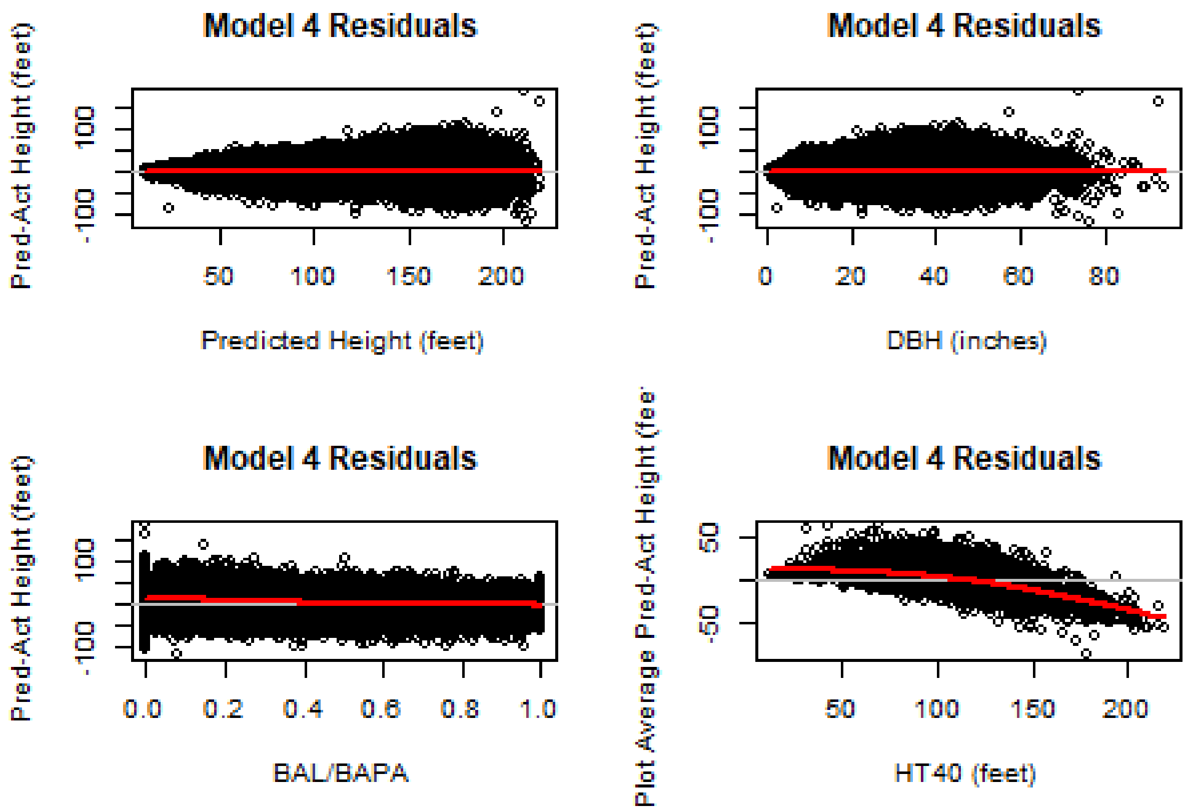


Model 4: Exponential Model Form (Schumacher 1939, see Curtis 1967, Equation 6)

$$HT = bh + exp(B_0 + B_1 * DBH^{B_2})$$

Table 16: HT-DBH Model 4 fit summary.

Model	A	B	C	RegSE	AIC
4	6.552	-5.611	-0.3667	19.57	1065989


Model 4 Residuals
Pred-Act Height (feet)
Predicted Height (feet)
Model 4 Residuals
Pred-Act Height (feet)
DBH (inches)
Model 4 Residuals
Pred-Act Height (feet)
BAL/BAPA
Model 4 Residuals
Plot Average Pred-Act Height (fee
HT40 (feet)

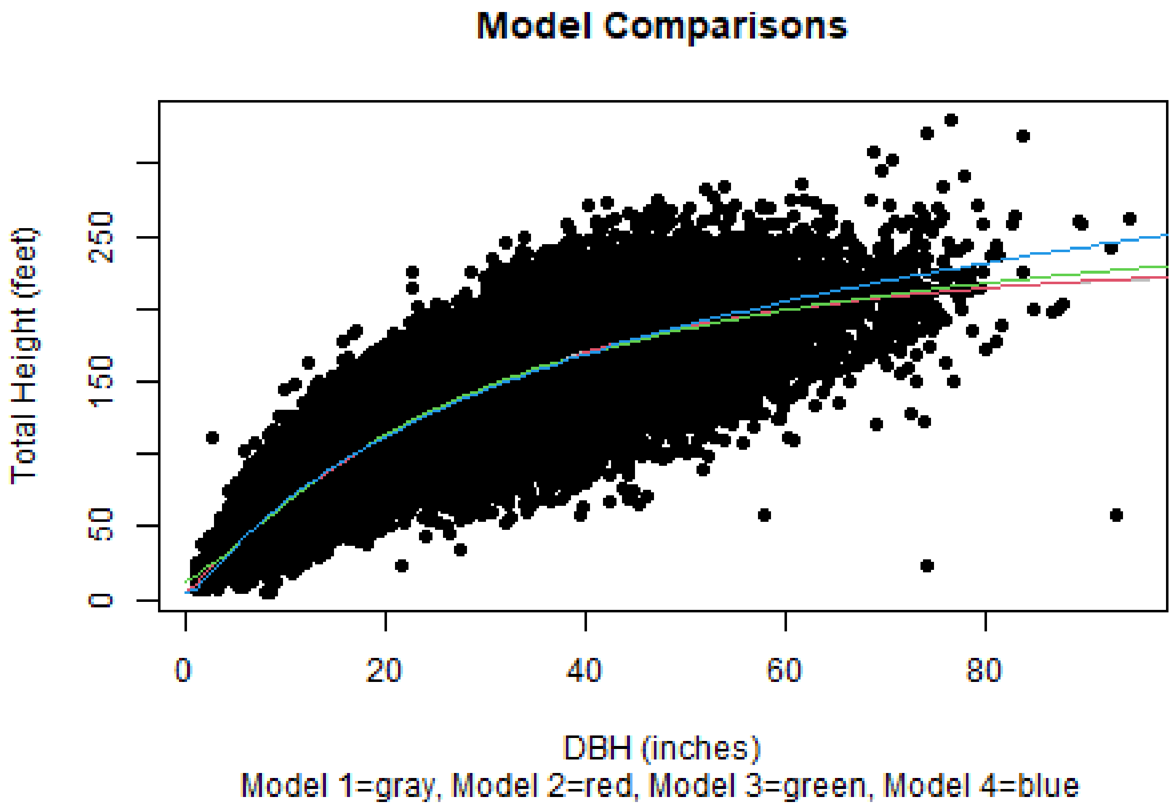


Table 17: HT-DBH Model fit comparisons.

Model	A	B	C	RegSE	AIC
1	225.4	-0.03055	1.021	19.52	1065477
2	226.3	-0.03333	1.029	19.52	1065475
3	5.671	-26.78	7.256	19.55	1065815
4	6.552	-5.611	-0.3667	19.57	1065989

The fits of the four models are very similar for the bulk of the data with small difference near DBH = 0 (model 3) and where they asyptote (model 4).

Temesgen and others (2007) identify model 4 has being unbiased and most precise for the most species they fit, although the differences are small. Martin and Flewelling (1998) reviewed several model forms (including model 4) and concluded that this was the best performer using inventory data for five western Washington species.

Zhang (1997) fit six nonlinear functions to 10 inter mountain species (including the four used here) and similar predictions for the bulk of the data but differences for the larger diameters in the data set (asymptotic differences) and that model 4 had a larger over-prediction for the biggest DBH class. Huang and Titus (1992) compared 20 equations on 9 species groups from Alberta (equations 13, 12, 18 and 10 represent the four models above) and recommended models 1, 2, and 3 over model 4 for general use.

6.0 Appendix – Mixed Effects Models Compared

This section compares the four models fit as mixed models with random effects on parameters B0 and B1, except for model 4 which has a single random effect on B0.

Model 1: Weibull Model Form

Table 18: Mixed Effects HT-DBH Model 1 fit summary.

Model	A	B	C	RegSE	AIC
1	145.1	-0.03711	1.248	10.58	954702

Model 2: Chapman-Richards Model Form

Table 19: HT-DBH Model 2 fit summary.

Model	A	B	C	RegSE	AIC
2	175.7	-0.05172	1.015	10.61	952121

Model 3: Exponential Model Form (see Wykoff 1986, page 12)

Table 20: HT-DBH Model 3 fit summary.

Model	A	B	C	RegSE	AIC
3	5.174	-13.03	2.581	10.59	950029

Model 4: Exponential Model Form (Schumacher 1939, see Curtis 1967, Equation 6)

Table 21: HT-DBH Model 4 fit summary.

Model	A	B	C	RegSE	AIC
4	5.302	-17.25	4.934	11.44	959373

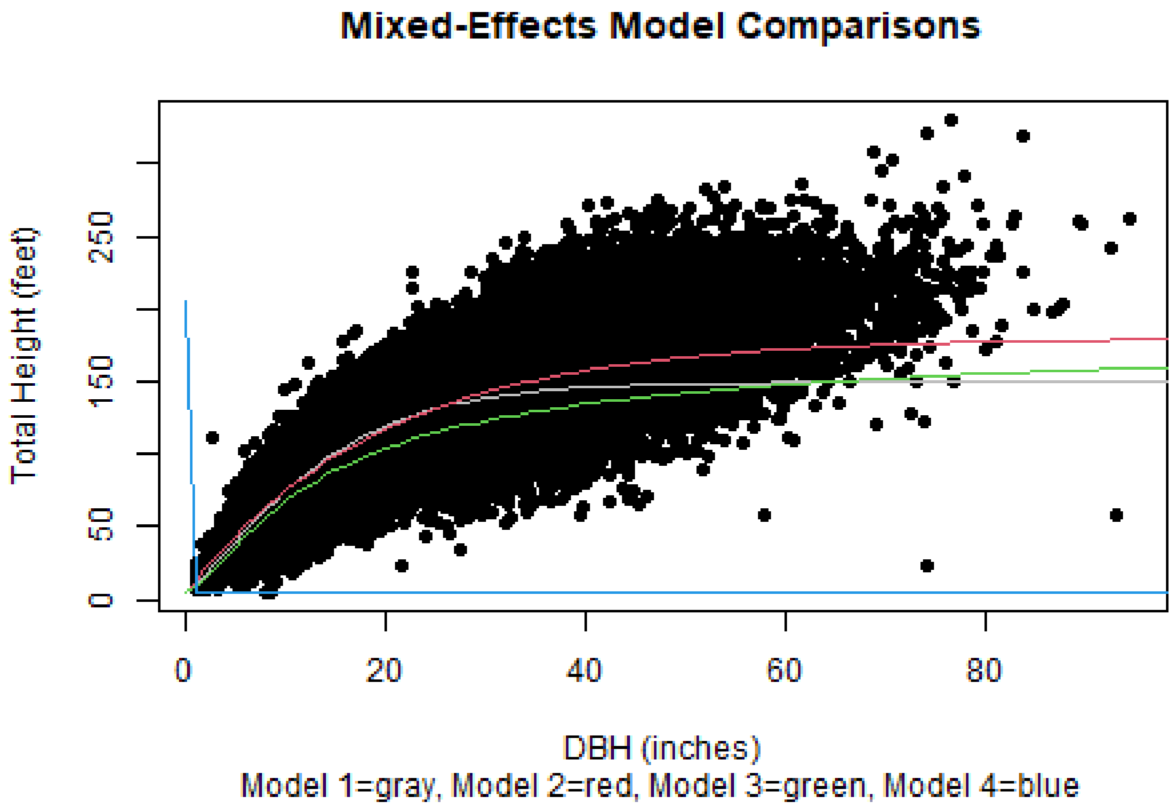


Table 22: HT-DBH Model fit comparisons.

Model	A	B	C	RegSE	AIC
1	145.1	-0.03711	1.248	10.58	954702
2	175.7	-0.05172	1.015	10.61	952121
3	5.174	-13.03	2.581	10.59	950029
4	5.302	-17.25	4.934	11.44	959373

Model 4 has 1 random effect of B0 and ther other have random effects on B0 and B1.